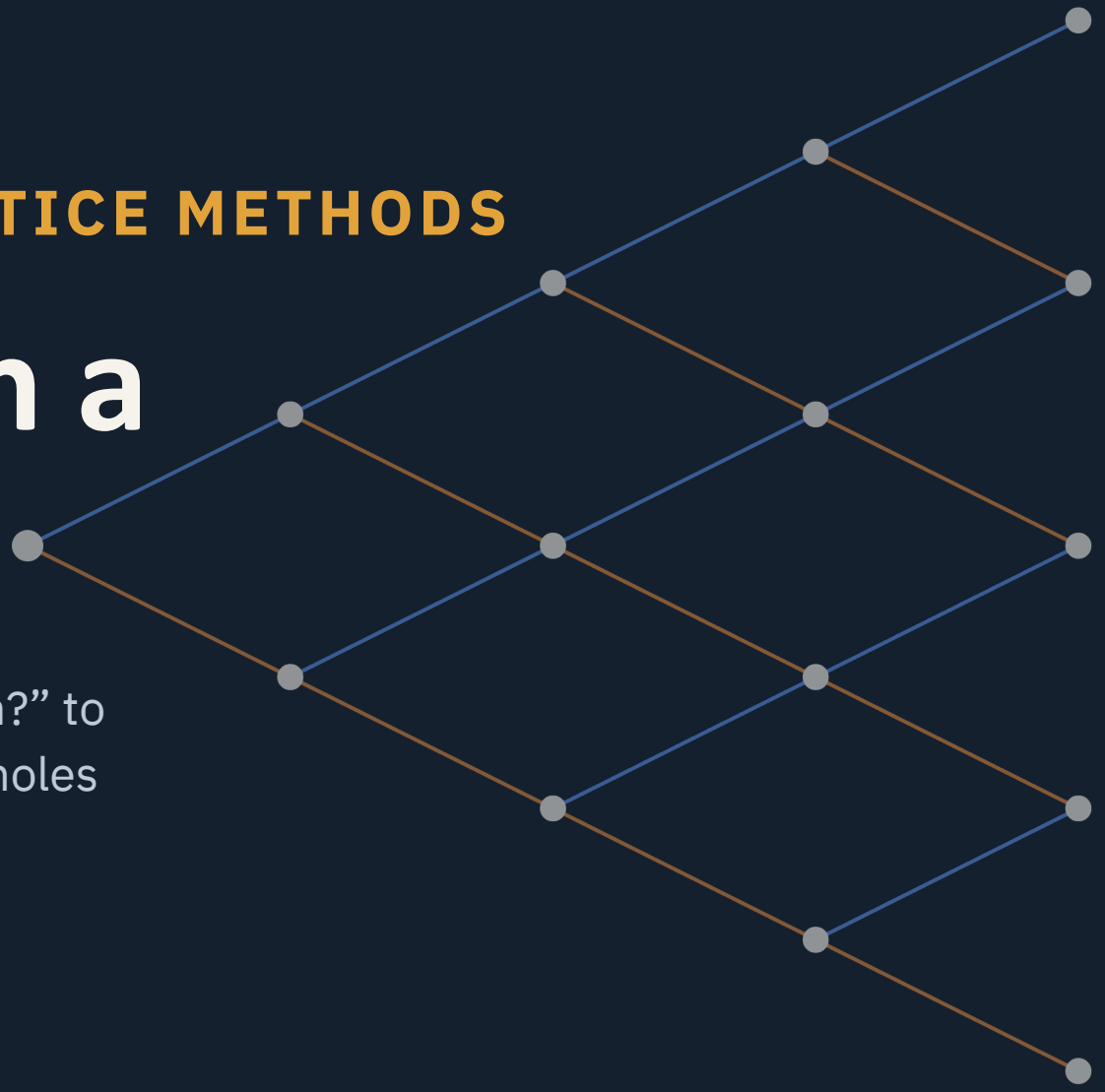
OPTIONS FOUNDATIONS · LATTICE METHODS

# Pricing options on a lattice



An intuitive walk from “what is this contract worth?” to the binomial tree — and why that tree is Black-Scholes in discrete form

J. Persson 2026

ROADMAP

# Six stops on the way to a price

| | | |
|---|---|---|
| 00 | **Options 101** | What an option is, what it is written on, why people use it — and how it has made and lost fortunes |
| 01 | **The pricing problem** | Payoffs, time value of money, why “expected payoff” fails, and the Black-Scholes answer |
| 02 | **One step** | Up with probability p, down with 1 – p — and why p drops out of the price |
| 03 | **The lattice** | Forward propagation of prices and probabilities; backward induction of values |
| 04 | **Using the lattice** | Hedging node by node, Greeks for free, American early exercise |
| 05 | **Back to Black-Scholes** | The backward step is a discretised PDE; the tree converges to the formula |

# Why should you care about options?

### In your pay slip

Employee stock options are call options on your employer's shares: the right to buy at a fixed price, later.

**15.2 bn**

### In the market

US option contracts cleared in 2025 — about 60 million a day, up 24% on 2024.

### In production code

Exchanges, brokers and banks price and risk-manage options all day — with exactly the models in this deck.

*An option is a programmable promise: a payoff function, a deadline — and a price someone has to compute.*

Source: OCC, annual volume report 2025 (published January 2026)

# Anatomy of an option: a right, not an obligation

```
struct OptionContract {
  underlying: "ACME shares"
  type:       CALL | PUT
  strike K:   100.00
  expiry T:   2027-03-19
  style:      EUROPEAN | AMERICAN
  premium:    5.20
}
// CALL: right to BUY at K
// PUT:  right to SELL at K
// EUROPEAN: exercise at T only
// AMERICAN: exercise any time up to T
// premium: paid today by the buyer
```

**Buyer (holder)**

Pays the premium, gets the right. Exercises only if it pays. Worst case: loses the premium.

**Seller (writer)**

Collects the premium, carries the obligation. Worst case: large losses.

**Call ≈ a price lock** (like a deposit reserving a house at today's price).
**Put ≈ an insurance policy** on what you own.

# The underlying: almost anything with a price

### Single stocks
Shares of one company. A US contract covers 100 shares.

### Indices & ETFs
S&P 500, Euro Stoxx 50, OMXS30 — a bet or hedge on a whole market.

### Currencies
EUR/USD, USD/SEK — exporters lock in exchange rates.

### Commodities
Oil, gold, grain — usually options on futures contracts.

### Interest rates
Caps, floors, swaptions — borrowers insure against rising rates.

### Volatility itself
VIX options — a bet on how much prices will move, not which way.

In this deck: one **non-dividend-paying stock** — the simplest underlying. Everything generalises.

2025 US cleared volume: single stocks 8.3 bn · ETFs 5.7 bn · indices 1.3 bn contracts (OCC)

# Why use options? Insurance, leverage, income

### Hedge

Own the stock, buy a put: your loss has a floor. A pension fund insuring its portfolio.

### Speculate

A small premium buys a large exposure: leverage, with the loss capped.

### Earn income

Sell calls on stock you own and collect premiums — giving away upside you didn't expect.

**Leverage in numbers** — stock at 100; a call with K = 100 costs 5:

| Stock moves to | Buy 1 share for 100 | Buy 1 call for 5 |
|---|---|---|
| 120 | +20 · +20% | +15 · +300% |
| 100 | 0 · 0% | −5 · −100% |
| 90 | −10 · −10% | −5 · −100% |

# Options are older than stock markets

## ~600 BC

### Thales' olive presses

Aristotle tells how Thales of Miletus paid small deposits for the use of olive presses before a bumper harvest — then rented them out at a profit. The first recorded call option.



## 1636–37

### Tulip mania

Dutch traders bought and sold the right to buy bulbs at fixed prices. When prices collapsed in February 1637, many contracts were never honoured in full.



## 1870s

### Russell Sage

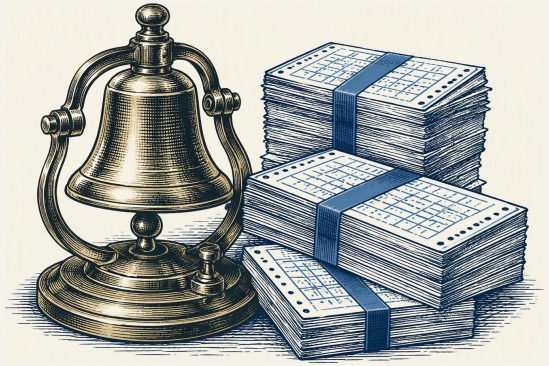
A Wall Street financier combined puts, calls and stock into synthetic loans — using the put-call parity we will check later.



## 1973

### CBOE opens

First exchange for standardised, centrally cleared stock options. Same year as the Black-Scholes paper. Today: ~60 million US contracts a day.



# …and they have blown up spectacularly

### 1987 · Black Monday

“Portfolio insurance” copied a put by selling stock as prices fell — the very hedge we’ll build. Everyone sold at once: the Dow fell 22.6% on 19 October.

### 1995 · Barings Bank

Trader Nick Leeson sold Nikkei options betting on calm markets. The Kobe earthquake hit; losses of £827m sank a 233-year-old bank.

### 1998 · LTCM

A hedge fund with Merton and Scholes on its board lost $4.6bn in under four months. The New York Fed brokered a $3.6bn rescue by 14 firms.

### 2021 · GameStop

Retail traders bought calls en masse. Dealers hedging them had to buy shares, pushing the price higher still — a “gamma squeeze”.

*Lesson: the model is a tool. Its assumptions are where the bodies are buried.*

# Black Monday: hedging into a falling market

**19 OCTOBER 1987**

## −22.6%

Dow Jones decline in one trading session

**BACKGROUND**

Portfolio insurance sought to mimic a protective put by cutting equity exposure as markets fell.

**FEEDBACK LOOP**

Simultaneous selling added pressure to falling prices. Trading systems and liquidity struggled under the volume.

**AFTERMATH**

The Fed supplied liquidity. Exchanges later introduced circuit breakers to pause rapid sell-offs.



*A hedge depends on the ability to trade when markets move.*

https://www.federalreservehistory.org/essays/stock-market-crash-of-1987
https://www.federalreserve.gov/pubs/feds/2007/200713/
https://www.nber.org/system/files/chapters/c10958/c10958.pdf

# Barings Bank: hidden trades, failed controls



**FEBRUARY 1995**

## £827m

Reported trading losses

**BACKGROUND**

Nick Leeson took unauthorised futures and options positions in Singapore while overseeing both trading and settlement.

**HIDDEN EXPOSURE**

He concealed losses in an error account. The Nikkei decline after the Kobe earthquake worsened the damage.

**OUTCOME**

Barings entered administration on 26 February. ING subsequently acquired the bank for £1.

*Independent controls matter as much as a pricing model.*

https://www.rba.gov.au/publications/bulletin/1995/nov/1.html
https://www.britannica.com/event/bankruptcy-of-Barings-Bank

# LTCM: leverage magnified model risk

**SEPTEMBER 1998**

## $3.6bn

Private rescue funded by 14 financial firms

**BACKGROUND**

Founded in 1994, Long-Term Capital Management used models to trade price differences between similar securities.

**STRESS**

High leverage amplified losses during the Asian and Russian financial crises. Turbulent markets made a forced sale dangerous.

**RESCUE**

The New York Fed convened creditors. Their capital injection allowed an orderly unwind without Fed money at risk.



*Modelled value cannot guarantee funding or market liquidity.*

https://www.federalreservehistory.org/essays/ltcm-near-failure
https://en.wikipedia.org/wiki/Long-Term_Capital_Management

# GameStop: a rally with disputed mechanics



*Gamma hedging is a mechanism, not proof of what drove a rally.*

JANUARY 2021

## Retail surge

Social attention met heavy short interest

**BACKGROUND**

Retail interest and social-media attention surged alongside GameStop's share price and trading volume.

**MARKET FRICTION**

Several brokers temporarily restricted trading. The episode raised questions about clearing, settlement and short selling.

**EVIDENCE CHECK**

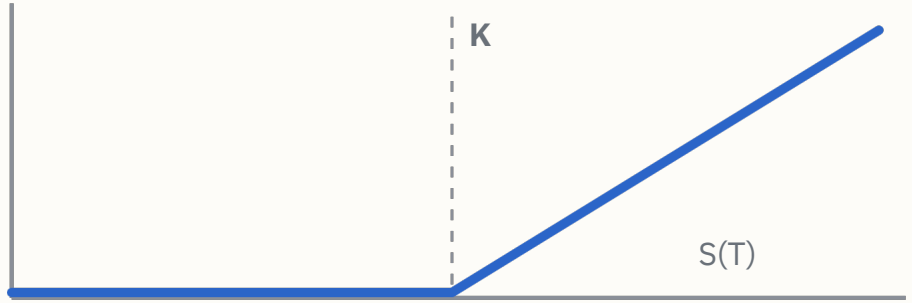
SEC staff found no evidence of a January gamma squeeze. Some academics dispute the analysis, so the explanation remains contested.

https://www.sec.gov/newsroom/press-releases/2021-212
https://www.sec.gov/files/staff-report-equity-options-market-struction-conditions-early-2021.pdf
https://clsbluesky.law.columbia.edu/2022/02/22/an-academic-critique-of-the-secs-gamestop-report/
https://www.shacknews.com/article/127195/sec-issues-gamestop-gme-report-on-what-happened-in-january-2021

# An option is a payoff function of a future price

## Call option

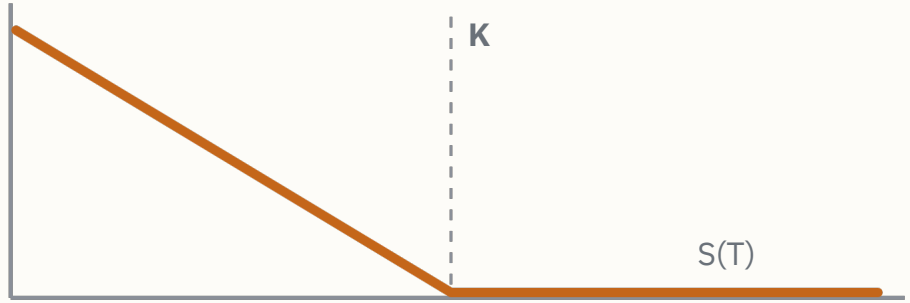
The right — not the obligation — to **buy** the stock at strike K on expiry date T.




payoff = max(S(T) − K, 0)

## Put option

The right — not the obligation — to **sell** the stock at strike K on expiry date T.




payoff = max(K − S(T), 0)

# Delta and gamma describe how option value moves

The payoff tells us what the option pays at expiry. Before expiry, its value V changes as the stock price S moves.

## Delta: how much does option value move?

Delta measures the local change in option value for a small change in the stock price.

With delta 0.50, a small $1 rise in the stock adds about $0.50 to the option's value.

## Gamma: how much does delta change?

Delta itself changes as the stock moves. Gamma measures how quickly it changes.

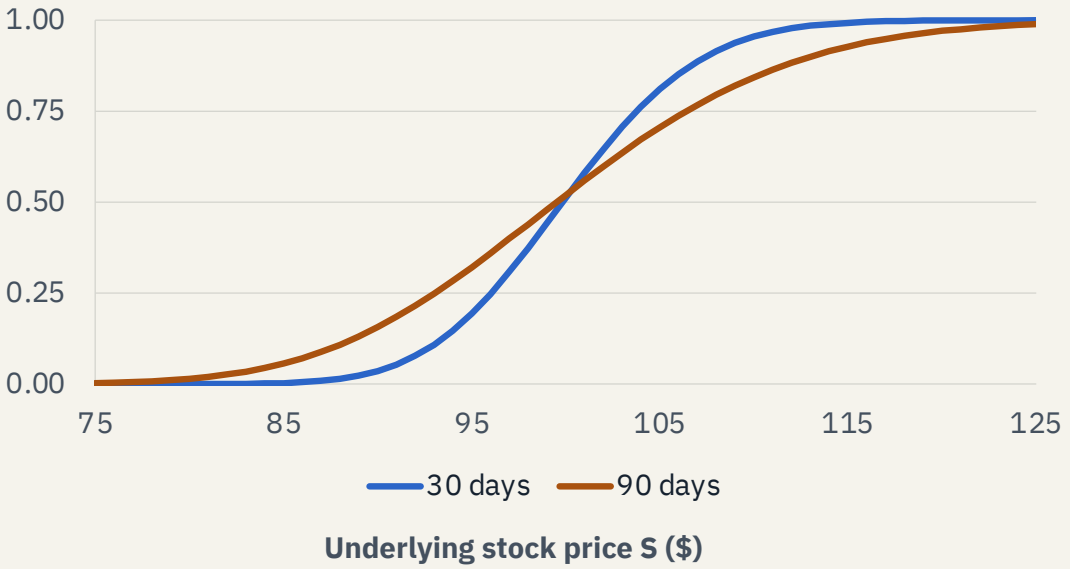
That changing sensitivity explains why a stock hedge needs rebalancing.

The next curves show delta and gamma for a call option before expiry.

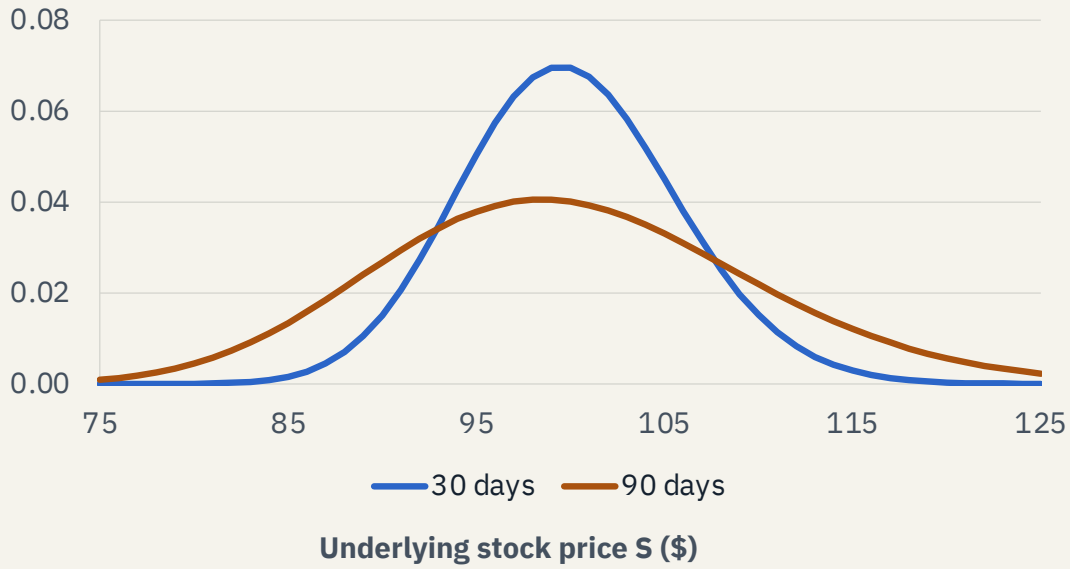
# Gamma measures how quickly delta changes

Delta $\Delta = \partial V/\partial S$ is price sensitivity. Gamma $\Gamma = \partial\Delta/\partial S = \partial^2 V/\partial S^2$ is its slope.

**CALL DELTA (shares per option share)**




**CALL GAMMA (delta change per $1)**




Stock can offset delta but has zero gamma. Gamma-neutral hedging adds options with offsetting curvature.

Illustrative Black–Scholes model: European calls, K = $100, volatility = 20%, r = q = 0, 365-day year.

# How to read the delta and gamma curves

## THE VARIABLES

### V · Option value

The call's current value in dollars per underlying share. A 100-share contract is worth 100 × V.

### S · Stock price

The current price of one underlying share, in dollars. Both figures use S on the horizontal axis.

### Δ · Delta = ∂V/∂S

The local slope of V versus S. With Δ = 0.50, a small $1 rise in S adds about $0.50 to V.

### Γ · Gamma = ∂Δ/∂S

The local slope of the delta curve, or curvature of V. Γ = 0.07 means about 0.07 more delta per $1 rise in S, locally.

## THE DELTA AND GAMMA FIGURES

### Left: delta rises from near 0 to near 1

Far below the $100 strike, the call reacts weakly to stock moves. Far above it, it behaves like one share. The 30-day curve rises more steeply near the strike.

### Right: gamma measures that steepness

Gamma is highest near the strike, where delta changes fastest. The blue 30-day peak is taller and narrower than the orange 90-day peak. Far from the strike, both approach zero.

### A snapshot with other inputs held fixed

Each curve varies only S. Blue means 30 days to expiry, orange 90 days. These are separate maturities, not a stock-price path through time. Short calls reverse the signs of both Greeks.

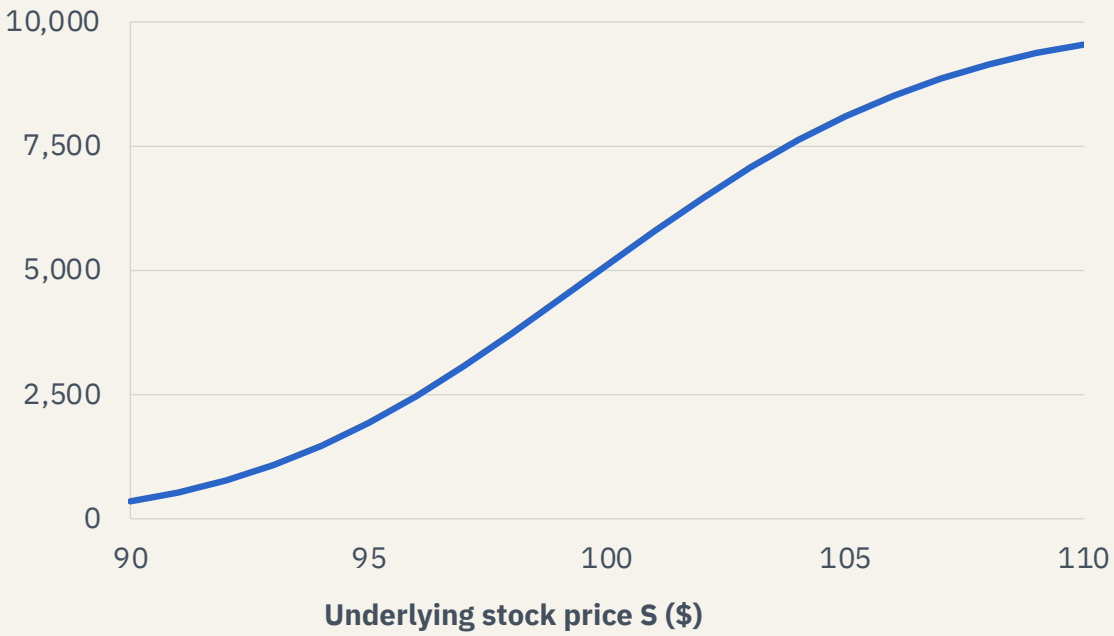
**For a small stock move: change in V ≈ Δ × change in S; change in Δ ≈ Γ × change in S.**

∂ means "change one input, hold the others fixed". Same assumptions as the figures: K = $100, volatility 20%, r = q = 0.

# A short-gamma stock hedge buys into a rise

Dealer is short 100 calls, each on 100 shares, with 30 days remaining. Stock hedge H = 10,000 × Δ.

**STOCK REQUIRED FOR DELTA NEUTRALITY**




| Stock | Call Δ | Hedge shares |
|---|---|---|
| $95 | 0.1933 | 1,932.71 |
| $100 | 0.5114 | 5,114.36 |
| $105 | 0.8105 | 8,104.59 |

**$100 to $105: buy 2,990.23 more shares.**
**$100 to $95: sell 3,181.65 shares.**

This rebalances delta while gamma remains negative. A long-gamma hedge trades in the opposite direction.

Same model as prior slide. Instantaneous scenarios, not a time path. Time and volatility fixed. Theoretical shares shown to two decimals.

# An options hedge can offset gamma locally

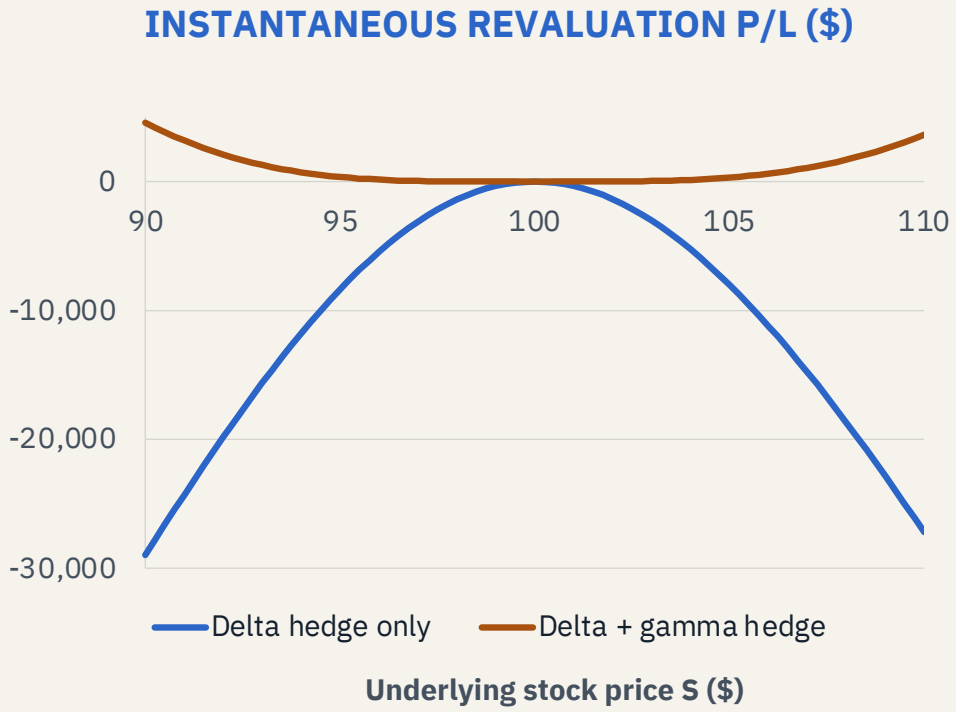
At S = $100: offset the short 30-day calls with 90-day calls, then use stock to remove the remaining delta.

$$n^B = -\Gamma^A / (100 \times \Gamma^B)$$
$$H = -(\Delta^A + n^B \times 100 \times \Delta^B)$$

A: signed original portfolio. B: one share of the 90-day call.

| Position | Quantity | Delta[1] | Gamma[2] |
|---|---|---|---|
| 30-day call | −100 | -5,114.36 | -695.48 |
| 90-day call | +173.347 | 9,010.64 | 695.48 |
| Stock | -3,896.28 | -3,896.28 | 0.00 |
| Total | | 0.00 | 0.00 |

Options: contracts × 100. [1] Delta: shares. [2] Gamma: shares/$.
Fractional hedge quantities. Totals use unrounded values.




Neutral only at the calibration point. Time decay, volatility changes, larger moves and trading costs still matter.

# The question: what is that payoff worth *today*?

**Inputs**

$S_0$ stock price now
K strike
T time to expiry
r risk-free rate
σ volatility



**Pricing model**

Black-Scholes formula,
or a binomial lattice



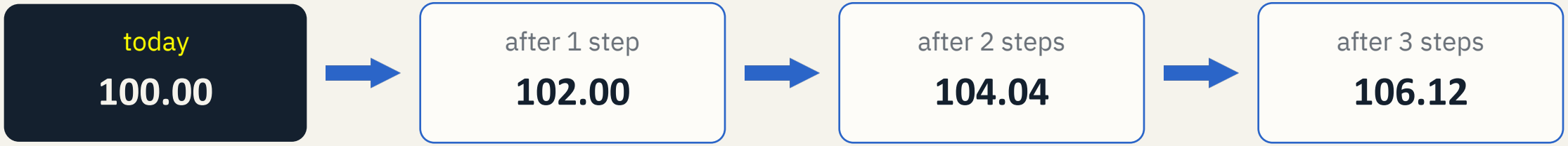
**Output**

$V_0$

one fair price today

The payoff rule is known exactly. The only unknown is the path the stock takes until T. A pricing model is a disciplined way of turning **uncertainty about the future** into **a single number now**.

# Time value of money: 100 today beats 100 next year




discounting = compounding run backwards: 106.12 in 3 steps is worth 100 today

**Compound (forward)**

$FV = PV \cdot R^n$

**R = 1.02 (<-> 2%)** per step: what a risk-free deposit or government bond pays.

**Discount (backward)**

$PV = FV / R^n$

In continuous time: **PV = FV · e^(−rT).** Same idea, smaller steps.

# Where the average comes from: probability-weighted payoffs

## A coin-flip game

Heads pays 10, tails pays 0, fair coin. Played many times, you win on average:

**0.5 · 10 + 0.5 · 0 = 5**

## In general: the expected value

$$E[X] = \Sigma\, p_i \cdot x_i$$

Each outcome $x_i$ weighted by its probability $p_i$. The weights sum to 1.

The developer's version: simulate many futures, average the payoffs (Monte Carlo).

```
N = 1_000_000
payoffs = [max(simulate_S_T() - K, 0) for _ in range(N)]
expected = sum(payoffs) / N
```

Law of large numbers: as **N** grows, the sample mean converges to **E[payoff].**

**The catch: the result is only as good as the probabilities inside simulate_S_T().**

# Put together: the “discounted expected payoff” recipe

Example: a stock at 100 will be at 110 or 90 one period from now. A call with K = 100. Risk-free growth R = 1.02.

| # | Step | Our example |
|---|---|---|
| 1 | List the possible futures | S = 110 or S = 90 |
| 2 | Payoff in each future | max(110−100,0)=10 · max(90−100,0)=0 |
| 3 | Assign probabilities (a forecast) | p = 0.7 up · 1 − p = 0.3 down |
| 4 | Average: expected payoff | 0.7·10 + 0.3·0 = 7.00 |
| 5 | Discount to today | 7.00 / 1.02 = 6.86 |

*Looks sensible … but .. soon we will see that at 6.86, someone else makes free money (and that we don't like!)*

# Why the recipe breaks: two inputs nobody agrees on

$$V_0 = E^p[\text{ payoff }] / (1 + \mu)^T \leftarrow ?$$

### Whose probability p?

The real chance of an up-move is unobservable. A bull and a bear would plug in different values.

### Which discount rate μ?

An option is riskier than the stock — and its riskiness changes as the price moves. No single rate fits.

### Many prices

Different opinions give different “fair” values. A market needs one price that everyone can agree on.

We will keep p in the model — and watch it disappear.

# Arbitrage locks in a gain without a market bet

The intuition: buy something cheaply and simultaneously sell an identical claim for more. The matching positions cancel the future risk.

**VENUE A: BUY ONE SHARE**

**$99**

**VENUE B: SELL THE SAME SHARE**

**$101**

**LOCKED-IN DIFFERENCE**

**$2**

The sale brings in $101 and the purchase costs $99, leaving $2.

The purchased share fulfils the sale's delivery obligation, so no stock-price exposure remains once both trades are locked in

Buying at $99 and hoping to sell at $101 tomorrow leaves price risk. That is speculation.

Formal idea: zero initial net investment, no possible loss, and a positive probability of profit.

*Illustrative quotes. Same security and settlement; simultaneous execution; no fees, taxes, funding or delivery risk.*

# No-arbitrage means matching payoffs have one price

No-arbitrage rules out a free-profit strategy. If two tradable portfolios deliver identical cash flows in every state, their prices today must match.

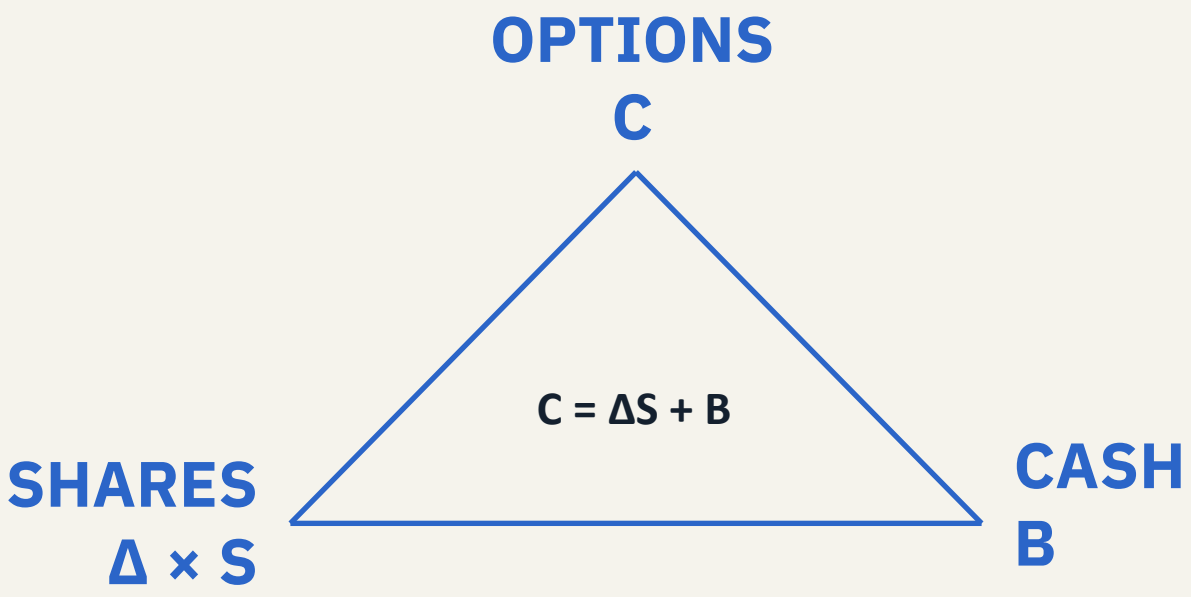
| Illustrative claim | Price today | Up-state payoff | Down-state payoff |
|---|---|---|---|
| A: cheaper claim | $5 | $10 | $0 |
| B: dearer claim | $6 | $10 | $0 |
| Buy A, sell B | Receive $1 | $0 net | $0 net |

Why the gap closes
Buying A and selling B puts upward pressure on A and downward pressure on B.

What must match
Every payoff, payment date, currency and credit risk. Equal expected payoffs are insufficient.

Illustrative frictionless example. Shorting and financing allowed. “No-arbitrage” is a consistency condition, not a forecast.

# The Options–Shares–Cash triangle




## THREE WAYS TO READ ONE IDENTITY

**1. Build the option**

$C = \Delta S + B$

Shares plus a bank position.

**2. Build the share position**

$\Delta S = C - B$

Option plus opposite cash.

**3. Build the cash position**

$B = C - \Delta S$

Option minus matching shares.

$B > 0$ means a deposit. $B < 0$ means borrowing.
$\Delta$ is the number of shares per option payoff unit.

Exact for the next slide's one-period, two-state model. Use signed, scaled positions; arbitrary holdings are not equivalent.

# Half a share and a loan reproduce the call

Same example as earlier: $S_0$ = \$100, next-period stock \$110 or \$90, call strike K = \$100, and cash grows by R = 1.02.

| Position per option share | Today | Stock at \$110 | Stock at \$90 |
|---|---|---|---|
| Buy 0.5 share | \$50.00 | \$55.00 | \$45.00 |
| Borrow \$44.12 | −\$44.12 | −\$45.00 | −\$45.00 |
| Shares + loan | \$5.88 | \$10.00 | \$0.00 |
| Call option | \$5.88 by no-arbitrage | \$10.00 | \$0.00 |

The other sides of the triangle
0.5 share − call = cash paying \$45.
2 calls + cash paying \$90 = 1 share.

**If the call costs \$6.86**
**Sell it and buy the \$5.88 replica.**
**Keep about \$0.98 today; future net payoff is \$0.**

Illustrative exact two-state model. No dividends or costs; borrowing and shorting allowed. Values round to cents.

# Why the triangle works: two outcomes, two controls

Think of the option as a pair of future payments: one if the stock rises, one if it falls. Shares and cash give us two ways to shape that pair.

**Shares set the difference**

A share pays more in the up state than in the down state. Changing the number of shares changes that gap. In our example, half a share pays $55 or $45: a $10 gap, just like the call's $10 or $0.

**Cash sets the level**

A bank deposit adds the same future amount in both states. A loan subtracts the same repayment. Subtract $45 from $55 and $45, and the two outcomes become exactly $10 and $0.

**Matching payments link prices**

Once the payments match in every state, the option and its replica do the same financial job. No-arbitrage links their prices. Rearranging the same positions gives the other sides of the triangle.

Reflection: if two packages pay exactly the same whatever happens, why pay more for one?

Exact in this one-period, two-state example. More trading dates require rebalancing; fixed shares and cash cannot match every option payoff.

# The way out: replicate the option, then invoke no-arbitrage

**The option**

Pays V↑ if the stock goes up, V↓ if it goes down

=

**Δ shares + B in the bank**

Chosen so it pays exactly V↑ and V↓ in the same two states

### Law of one price

Same payoff in every state ⇒ same price today.

### Otherwise: free money

Buy the cheap one, sell the dear one, collect the gap with zero risk.

### No opinions needed

The price of shares and cash is observable today. So is the option’s.

# A short history: continuous first, discrete second

**1900**

**Bachelier**

Thesis “Théorie de la spéculation” models prices as a random walk

**1965**

**Samuelson**

Geometric Brownian motion: returns, not prices, are random

**1973**

**Black, Scholes, Merton**

Hedging argument gives a PDE and a closed-form price

**1979**

**Cox, Ross, Rubinstein**

“A simplified approach”: the binomial lattice

**1997**

**Nobel Prize**

Awarded to Merton and Scholes (Black died in 1995)

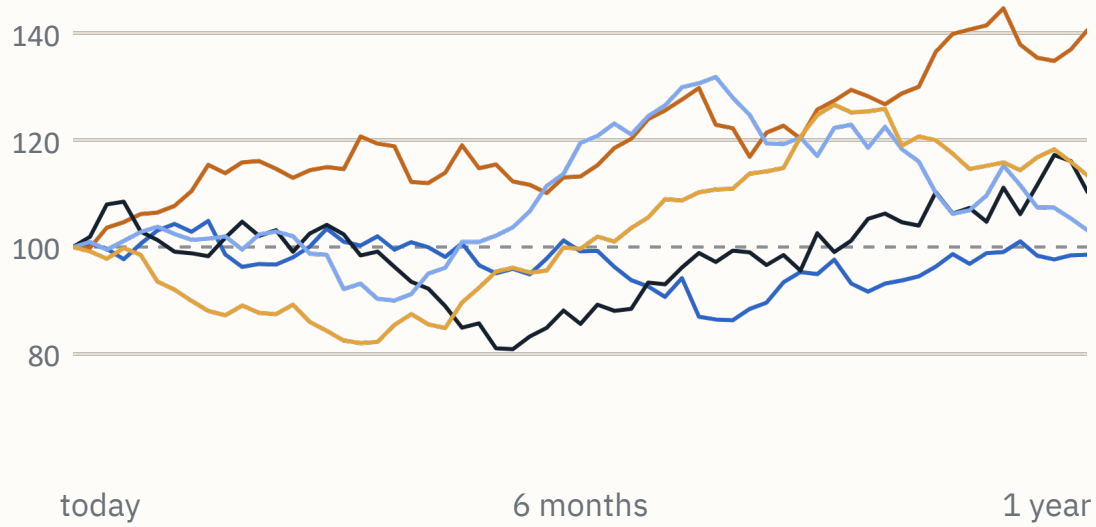
The lattice came **after** the formula — as a way to make the same economics visible, teachable and computable for contracts the formula cannot handle.

# First, the cast of characters

| Symbol | Meaning | Example |
|---|---|---|
| S | Stock price — it changes over time | 100 |
| t, T | Current time, and the expiry date (in years) | 0, 1 |
| K | Strike price written in the contract | 100 |
| r | Risk-free interest rate per year (continuous compounding) | 5% |
| σ sigma | Volatility: standard deviation of the stock's yearly return | 20% |
| μ mu | Expected return of the stock — watch it disappear | 8% |
| V(S, t) | The unknown: option value for any price S at any time t | ? |

We are not looking for one number but for a **function** V(S, t). Today's price is just one call: V(100, 0).

# How the stock moves: a steady drift plus random noise




**5 simulated paths · $S_0$ = 100 · μ = 8% · σ = 20%**

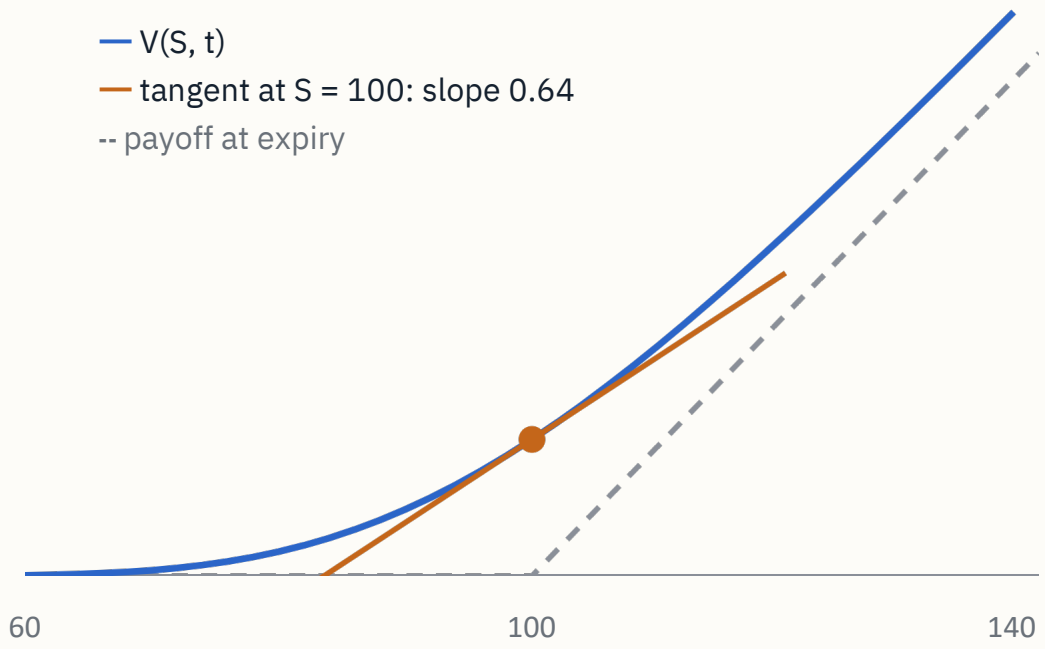
$$dS/S = \mu \cdot dt + \sigma \cdot dW$$

**dt** a tiny slice of time. **dS/S** the percentage price change in that slice.

**μ·dt** the steady drift: the expected growth.

**σ·dW** the noise: dW is a random shock with variance dt — countless tiny coin flips. Our lattice's up/down steps, made infinitely small.

# Reading ∂: how V reacts to one input at a time




**∂V/∂S · delta · slope**

(V(S+h,t) − V(S−h,t)) / 2h

**∂²V/∂S² · gamma · curvature**

(V(S+h) − 2V(S) + V(S−h)) / h²

**∂V/∂t · theta · time decay**

(V(S, t+h) − V(S, t)) / h

∂ = “change one argument, freeze the others”. A unit test with a tiny h computes each one.

# The Black-Scholes equation, one term at a time

$$\partial V/\partial t + r \cdot S \cdot \partial V/\partial S + \tfrac{1}{2}\sigma^2 S^2 \cdot \partial^2 V/\partial S^2 = r \cdot V$$

### Time decay

How the value changes as the clock ticks, price frozen (theta).

### Drift at r

Stock grows at the risk-free rate r, times the slope (delta). Not μ!

### Volatility × curvature

Randomness adds value where V is curved (gamma). No curve, no gain.

### = bank growth

Together the hedged option must grow like cash at rate r.

### Where it comes from

Hold the option, short $\Delta = \partial V/\partial S$ shares. The noise cancels; a riskless position must earn r. (The triangle slide, in continuous time.)

### What pins it down

$V(S, T) = \max(S - K, 0)$

The payoff at expiry. Solve backward in time from there.

# Meanwhile, in physics: the heat equation

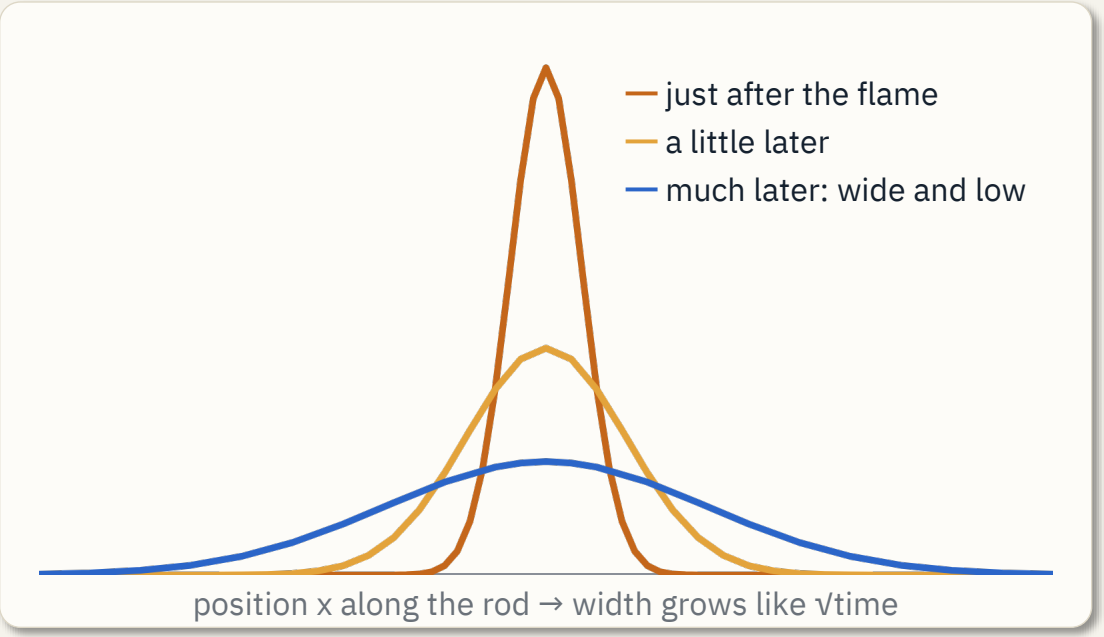


A metal rod, cold everywhere except one hot spot

$$\partial u/\partial \tau = \kappa \cdot \partial^2 u/\partial x^2$$

**u** temperature · **x** position on the rod · **τ** time elapsed · **κ** how fast heat conducts

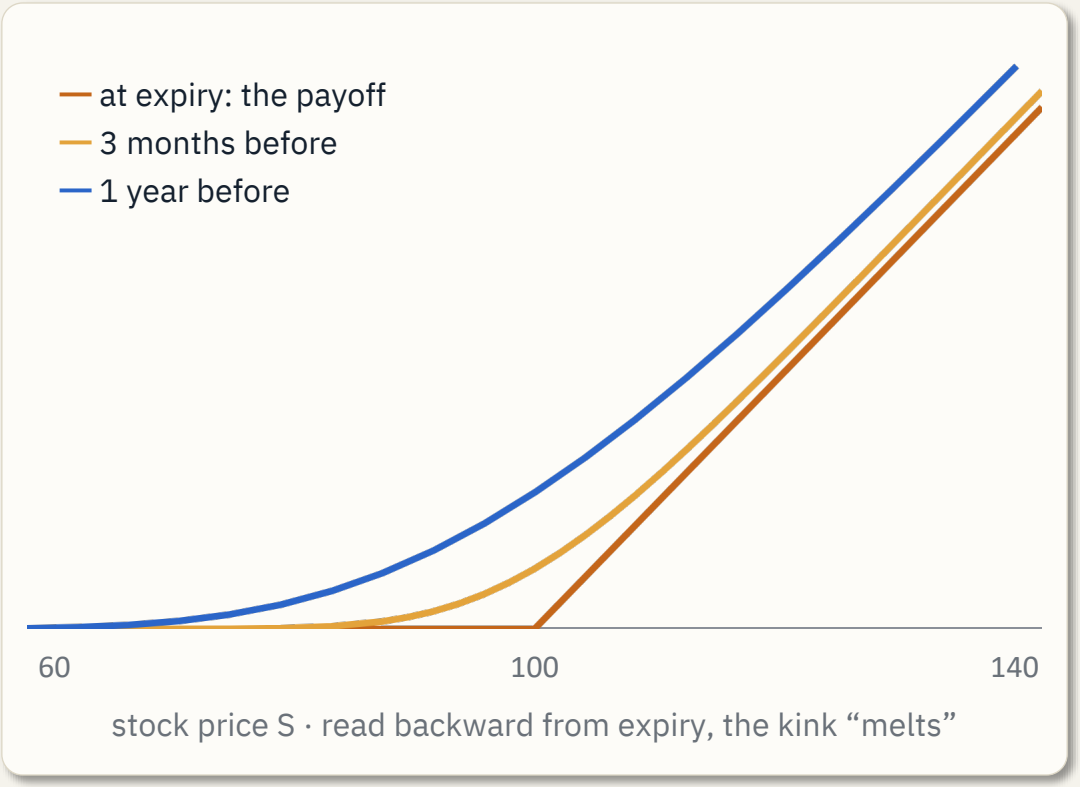
In words: a point warms up if it is colder than the average of its neighbours. The rate is the **curvature** — the same ∂² as gamma.




Fischer Black was trained in physics (Harvard BA 1959, PhD in applied maths 1964). In their 1973 paper, Black and Scholes showed that a change of variables turns their equation into the heat equation — whose solution was textbook material. Bachelier had already linked price randomness to Fourier's heat theory in 1900.

# Why heat? Both are random walks

| Heat in a rod | Option value |
|---|---|
| Molecules kick at random; spread grows like √time | Trades kick the price; spread grows like σ√T |
| Initial temperature profile | Payoff at expiry |
| Diffusion smooths sharp edges | Time to expiry smooths the kink |
| Conductivity κ | ½σ² |
| Grid step: average the neighbours | Lattice step: [q·V↑ + (1−q)·V↓] / R |



# Refactor: three substitutions reveal the heat equation

| | | |
|---|---|---|
| **1** | $\tau = T - t$ | Count **time to expiry**. The payoff becomes a starting condition, like the initial temperature. |
| **2** | $x = \ln S + (r - \tfrac{1}{2}\sigma^2)\cdot\tau$ | Use **log-price**: % moves become plain additive steps. The shift absorbs the drift term. |
| **3** | $V = e^{(-r\cdot\tau)} \cdot u$ | Pull out the **discounting**. The $r\cdot V$ term disappears. |

$$\partial u/\partial \tau = \tfrac{1}{2}\sigma^2 \cdot \partial^2 u/\partial x^2$$

start: $u(x, 0) = \max(e^x - K, 0)$

The heat equation with $\kappa = \tfrac{1}{2}\sigma^2$, and the payoff as the initial temperature. Like refactoring until your problem matches a library you already have.

# A call pays only when the final price exceeds the strike

A European call gives you the right to buy a share for a fixed price, the strike K, at expiry. With K = 100, the cash value at expiry is:

| Share price at expiry | 80 | 100 | 120 | 140 |
|---|---|---|---|---|
| Call payoff | 0 | 0 | 20 | 40 |

payoff = max(final_share_price − 100, 0)

Before expiry, we do not know which payoff will occur. Pricing combines the possible payoffs using specific weights, then converts that average to today's money.

**The next slide explains where those pricing weights come from.**

# “Risk-neutral” means weights chosen for pricing

A preview of the one-step example: a share costs 100 today and ends at 90 or 110. Over the same step, a risk-free bank account grows 100 to 102.

## The pricing weights

40% on 90 and 60% on 110

$0.40 \times 90 + 0.60 \times 110 = 102$

## The call with strike 100

Payoffs: 0 if down, 10 if up

$0.40 \times 0 + 0.60 \times 10 = 6$

Call price today = 6 / 1.02 = 5.88

The weights make the average share value grow at the bank rate. They need not match real-world odds. A matching share-and-cash portfolio justifies this price under the model’s assumptions.

# The heat equation performs repeated weighted averages

Slide 37 changed the coordinates to log-price and removed discounting. In those coordinates, solving the heat equation means averaging nearby payoff values.

## One illustrative averaging step, before discounting

| Nearby payoff entry | Left | Centre | Right |
|---|---|---|---|
| Value | 0 | 0 | 10 |
| Weight | 25% | 50% | 25% |

New centre value = 0.25 × 0 + 0.50 × 0 + 0.25 × 10 = 2.50

Repeat with many tiny steps: the averaging weights approach a bell curve. Its spread is $\sigma\sqrt{T}$ in log-price coordinates. More time or volatility spreads the weights farther.

# Black-Scholes calculates the whole average at once

For a European call on a stock with no dividends:

$$C = S \times N(d_1) - K \times \exp(-rT) \times N(d_2)$$

S = share price now  K = strike  r = annual risk-free rate

T = years to expiry  σ = annual volatility  C = call price today

N(z) adds up standard-normal weights below z. For example, N(0) = 50%. In code: norm.cdf(z).

$N(d_2)$ gives the pricing probability of finishing above K. $N(d_1)$ also accounts for higher share values in those outcomes.

$S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$

$d_1 = 0.35$, $d_2 = 0.15$, $N(d_1) \approx 0.6368$, $N(d_2) \approx 0.5596$

**Today's value received 63.6831 – strike payment 53.2325 = 10.4506**

# Why a lattice? Because it is the same idea, made discrete

### Discretise

Chop [0, T] into n steps of Δt. In each step the price can only go up by a factor u or down by a factor d.

### Dynamic programming

A node's value depends only on its two children. Fill a triangle backward: $O(n^2)$ time, $O(n)$ memory.

### Flexible

Change the rule at a node to handle early exercise, barriers or dividends — no new formula needed.

And as $n \to \infty$, the lattice price converges to the Black-Scholes-Merton price

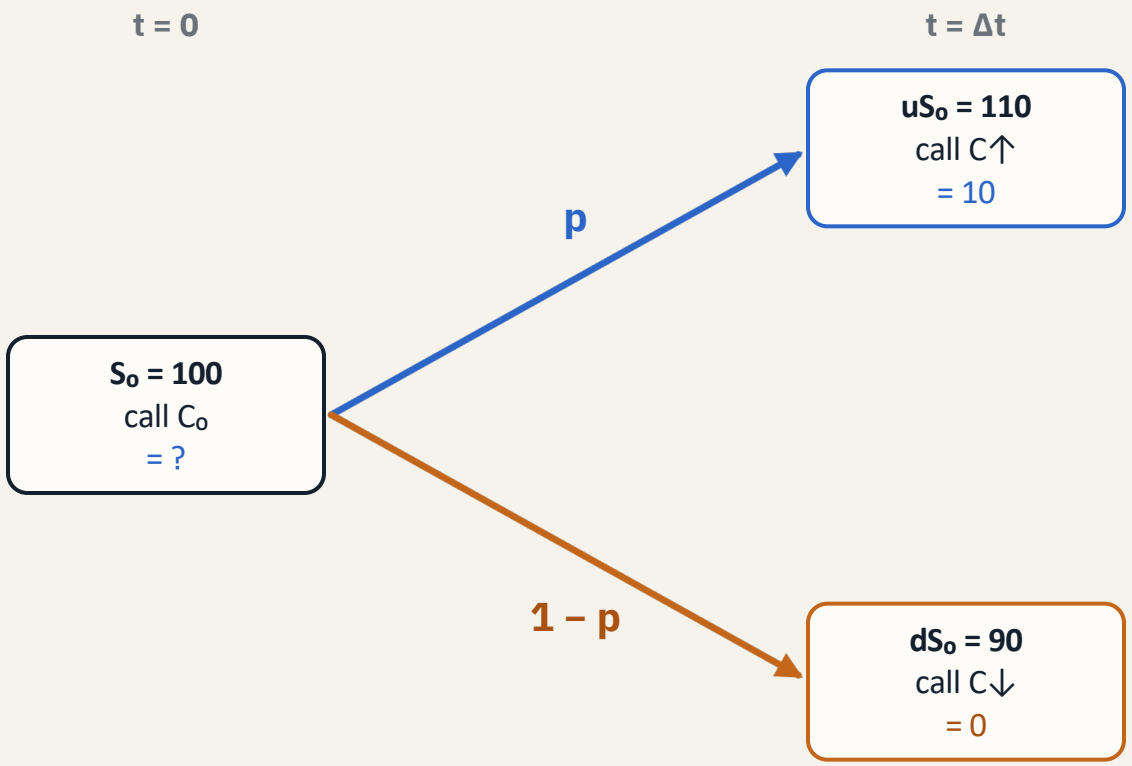
# The smallest possible world: one step, two states

Stock today **$S_0 = 100$**. After one step it is either

$u \cdot S_0 = 110$ prob p
$d \cdot S_0 = 90$ prob 1 – p

Cash grows by **$R = e^{(r\Delta t)} = 1.02$** per step, risk-free.

A call with **$K = 100$** pays 10 or 0. What is it worth now?

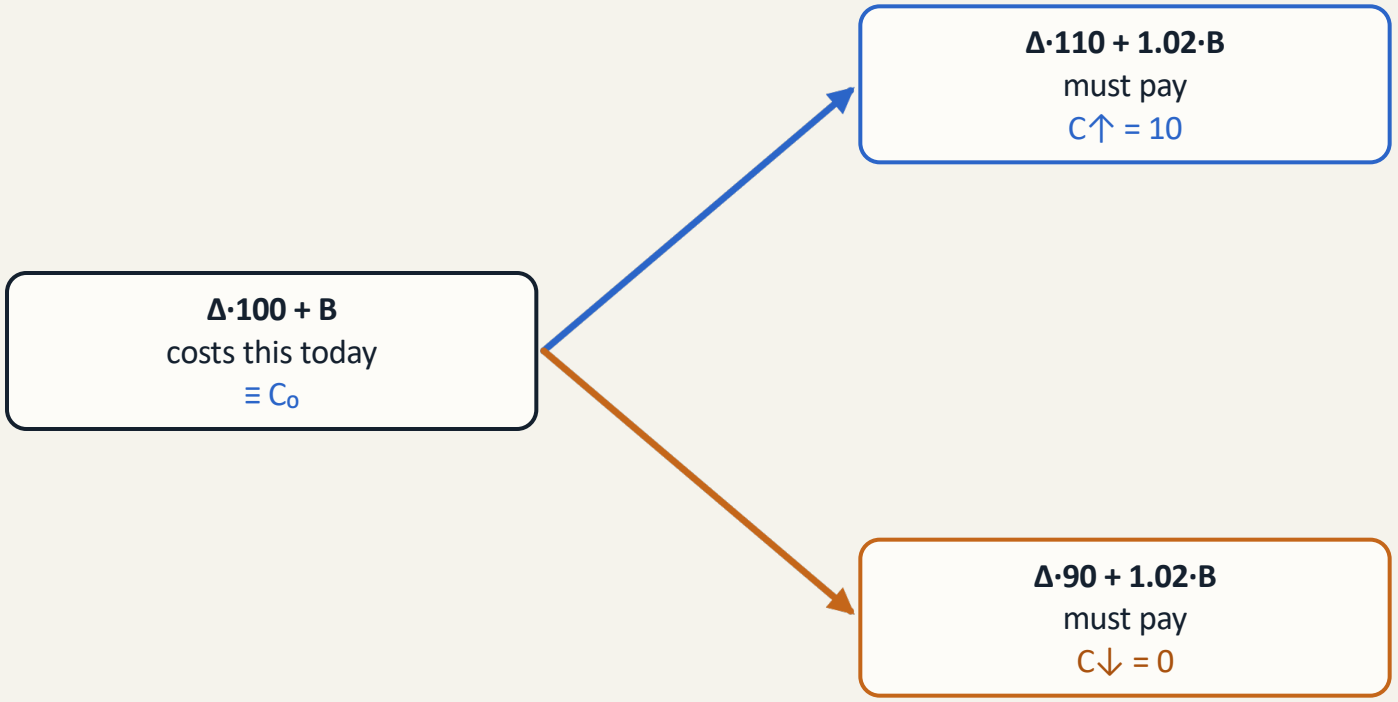


# Build a clone: Δ shares plus B in cash

Two unknowns, Δ and B. Two states, so two equations — one per branch.

$$\Delta = (C\uparrow - C\downarrow)/(S\uparrow - S\downarrow)$$
$$B = (C\downarrow - \Delta \cdot S\downarrow) / R$$

Δ is the slope of the option payoff across the two states: the **hedge ratio**.



# Why a clone exists: two points define a line

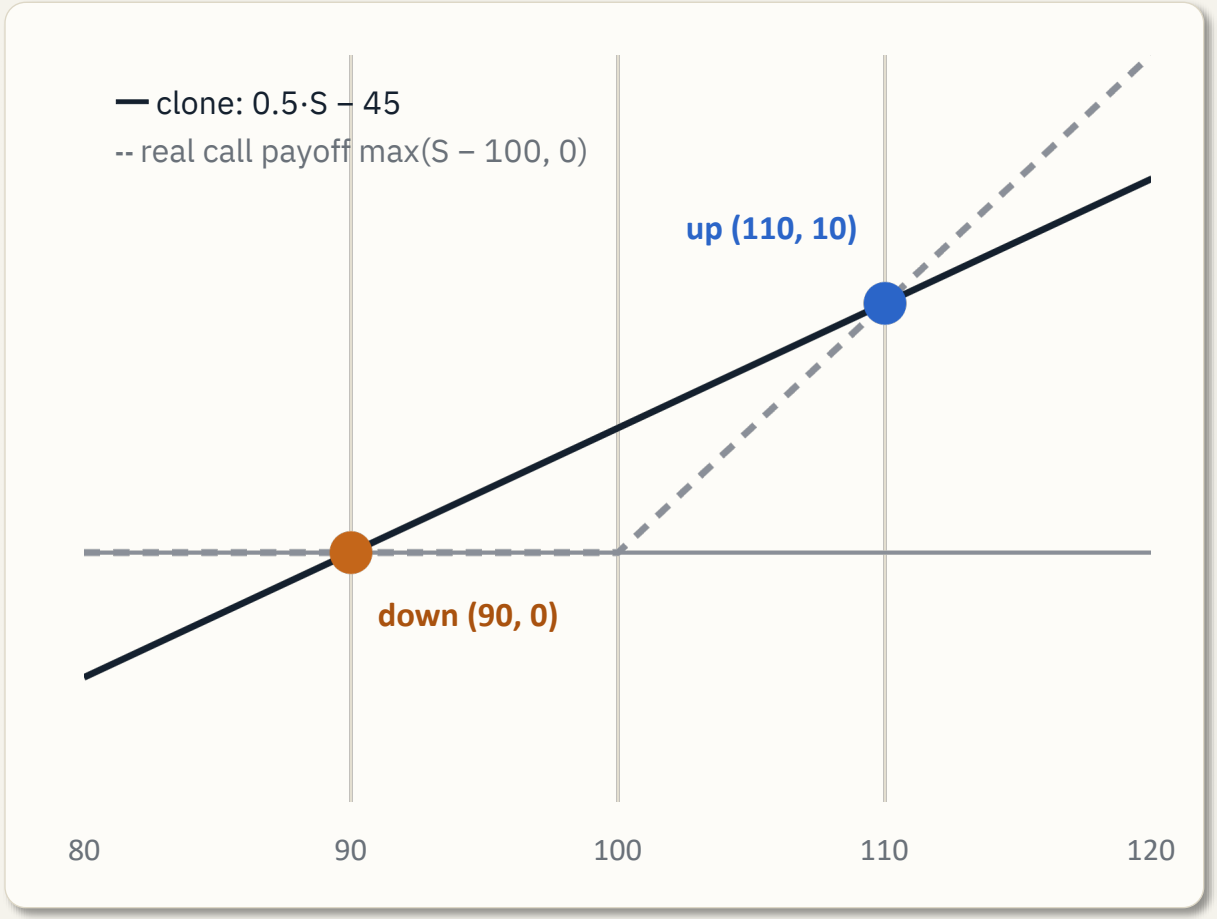
Plot value at the end of the step against the stock price S.

**Shares** give a value proportional to S: the **slope**. **Cash** gives the same amount whatever S does: the **level**.

$$\text{clone}(S) = \Delta \cdot S + R \cdot B$$

The option only has to match at two points. Exactly one line passes through both:

slope $\Delta = 10/20 = 0.5$
level $R \cdot B = -45 \rightarrow B = -44.12$



# Solve it: the clone costs 5.88, so the option does too

SHARES

**0.50**

$\Delta = (10 - 0)/(110 - 90)$

CASH (BORROWED)

**−44.12**

$B = (0 - 0.5 \cdot 90)/1.02$

OPTION PRICE

**5.88**

$C_0 = 0.5 \cdot 100 - 44.12$

| Check | Shares | Repay loan | Portfolio |
|---|---|---|---|
| Up (S = 110) | 0.5 × 110 = 55.00 | −44.12 × 1.02 = −45 | 10 = C↑ ✓ |
| Down (S = 90) | 0.5 × 90 = 45.00 | −44.12 × 1.02 = −45 | 0 = C↓ ✓ |

# If the price is wrong, a money machine appears

Suppose someone quotes the option at **6.50** instead of 5.88. Sell it, and buy the clone:

| Trade | Today | Up: S = 110 | Down: S = 90 |
|---|---|---|---|
| Sell 1 option | +6.50 | −10.00 | 0.00 |
| Buy 0.5 share | −50.00 | +55.00 | +45.00 |
| Borrow cash | +44.12 | −45.00 | −45.00 |
| Net | +0.62 | 0.00 | 0.00 |

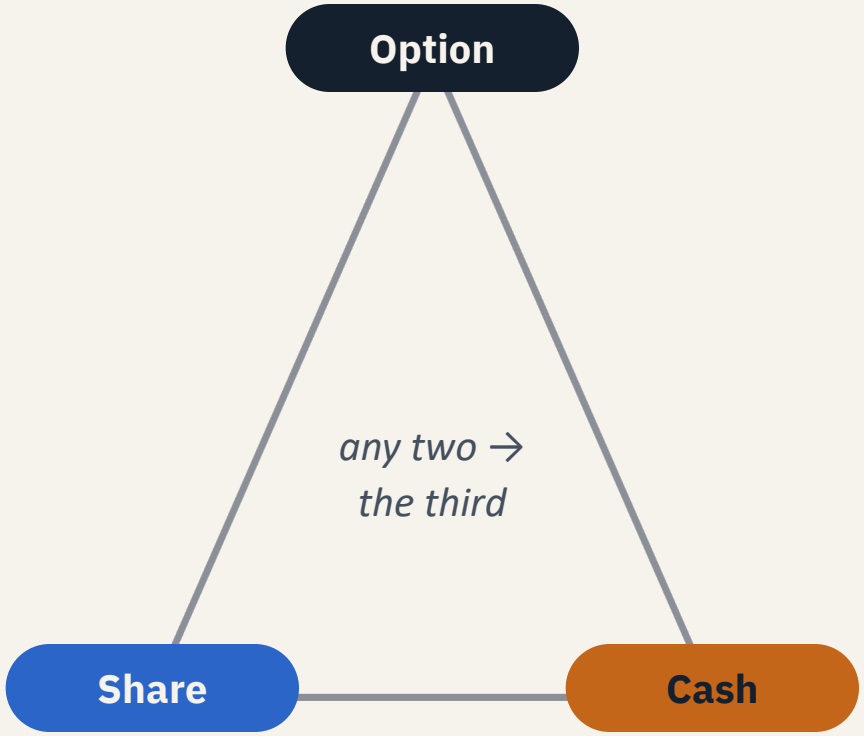
### Profit now, no risk later

Whatever the stock does, the clone pays off the option exactly. The 0.62 is free.

### Too cheap? Run it in reverse

At 5.00: buy the option, short the clone, pocket 0.88. Traders' code does this until only 5.88 is left.

# Share, option, cash: any two build the third




**Option = 0.5 share – 44.12 cash**

Replicate. Up: 55 – 45 = 10 ✓ Down: 45 – 45 = 0 ✓ Cost 5.88.

**Cash = 0.5 share – 1 option**

Hedge. Pays 45 in both states: riskless, so it must be worth 45/1.02 = 44.12. Hence option = 50 – 44.12 = 5.88.

**Share = 2 options + 88.24 cash**

Synthetic share. Up: 20 + 90 = 110 ✓ Down: 0 + 90 = 90 ✓ Cost 2·5.88 + 88.24 = 100 ✓

**The key move:** a delta-hedged option is a bond. A bond earns the risk-free rate. That alone fixes the price.

# Rearrange the algebra and a new probability appears

$$C_0 = \Delta \cdot S_0 + B = [\ (R - d)/(u - d) \cdot C\uparrow + (u - R)/(u - d) \cdot C\downarrow\ ] / R$$

$$q = (R - d) / (u - d)$$

$$C_0 = [\ q \cdot C\uparrow + (1 - q) \cdot C\downarrow\ ] / R$$

Example

$q = 0.12 / 0.20 = 0.60$

$C_0 = 0.6 \cdot 10 / 1.02 = 5.88$ ✓

*The real-world probability p appears nowhere.*

# Two worlds: p forecasts, q prices

| | Real world, p = 0.70 | Risk-neutral, q = 0.60 |
|---|---|---|
| Where it comes from | Opinion, history, research | (R − d)/(u − d): market data |
| Expected stock growth / step | 0.7·1.1 + 0.3·0.9 = 1.04 | 0.6·1.1 + 0.4·0.9 = 1.02 = R |
| Discounted E[call payoff] | 0.7·10 / 1.02 = 6.86 | 0.6·10 / 1.02 = 5.88 |
| Is that a tradeable price? | No: sell at 6.86, buy clone, keep 0.98 | Yes: the only arbitrage-free price |

### Intuition

Price as if every asset grows at the risk-free rate. The hedge removes risk, so risk preferences can't matter.

### Sanity condition

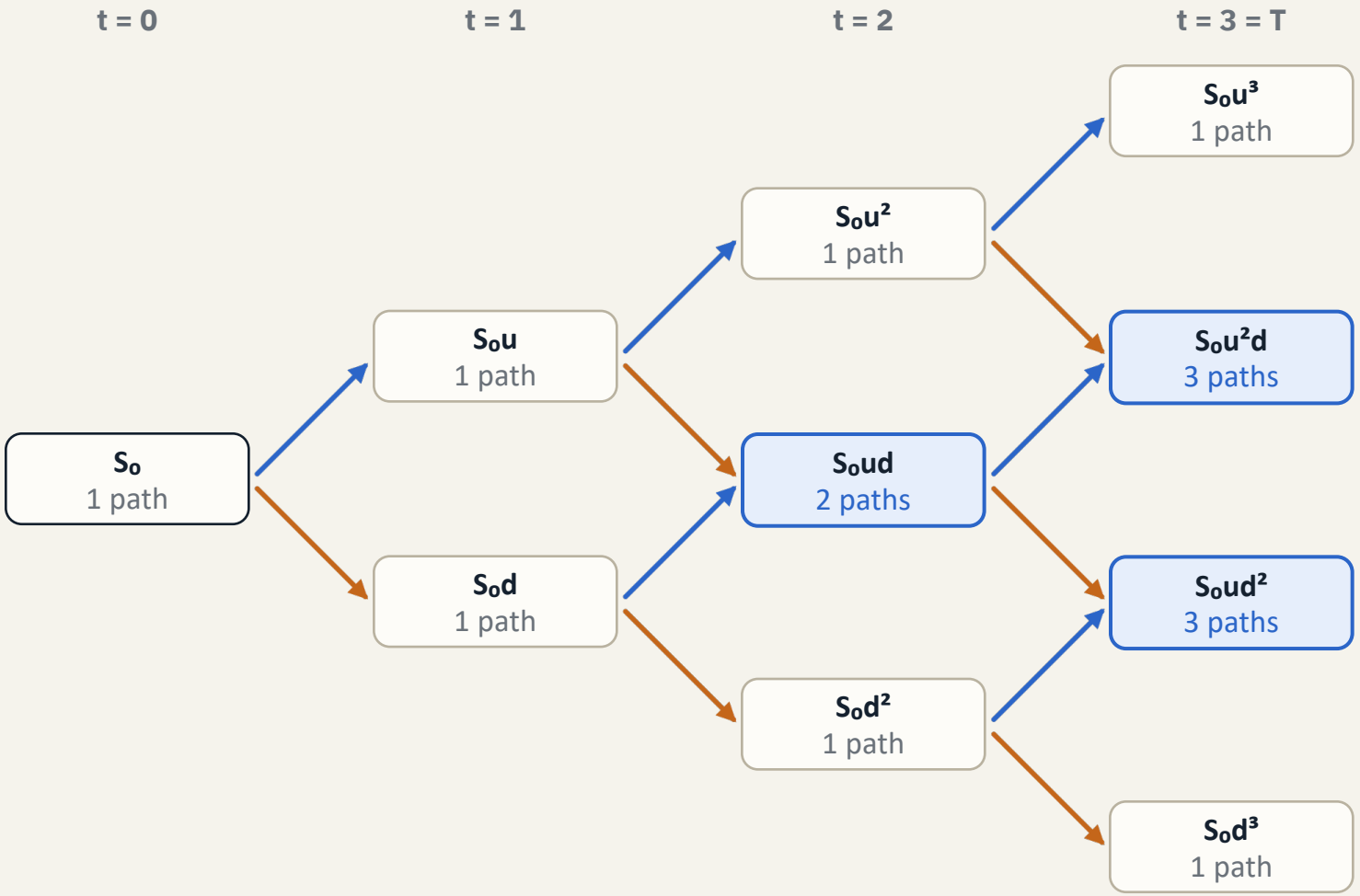
**d < R < u** ⇔ 0 < q < 1. Otherwise the stock or the bank dominates, an arbitrage.

# Stack the step: a recombining lattice

Repeat the one-step move n times. Because **u·d = d·u**, paths merge: up-then-down lands where down-then-up does.

node (i, j):
$S = S_0 \cdot u^j \cdot d^{i-j}$

i = step, j = number of up-moves. n steps give **n + 1 leaves**, not $2^n$ paths. With n = 100: 101 nodes instead of ~$10^{30}$ paths.

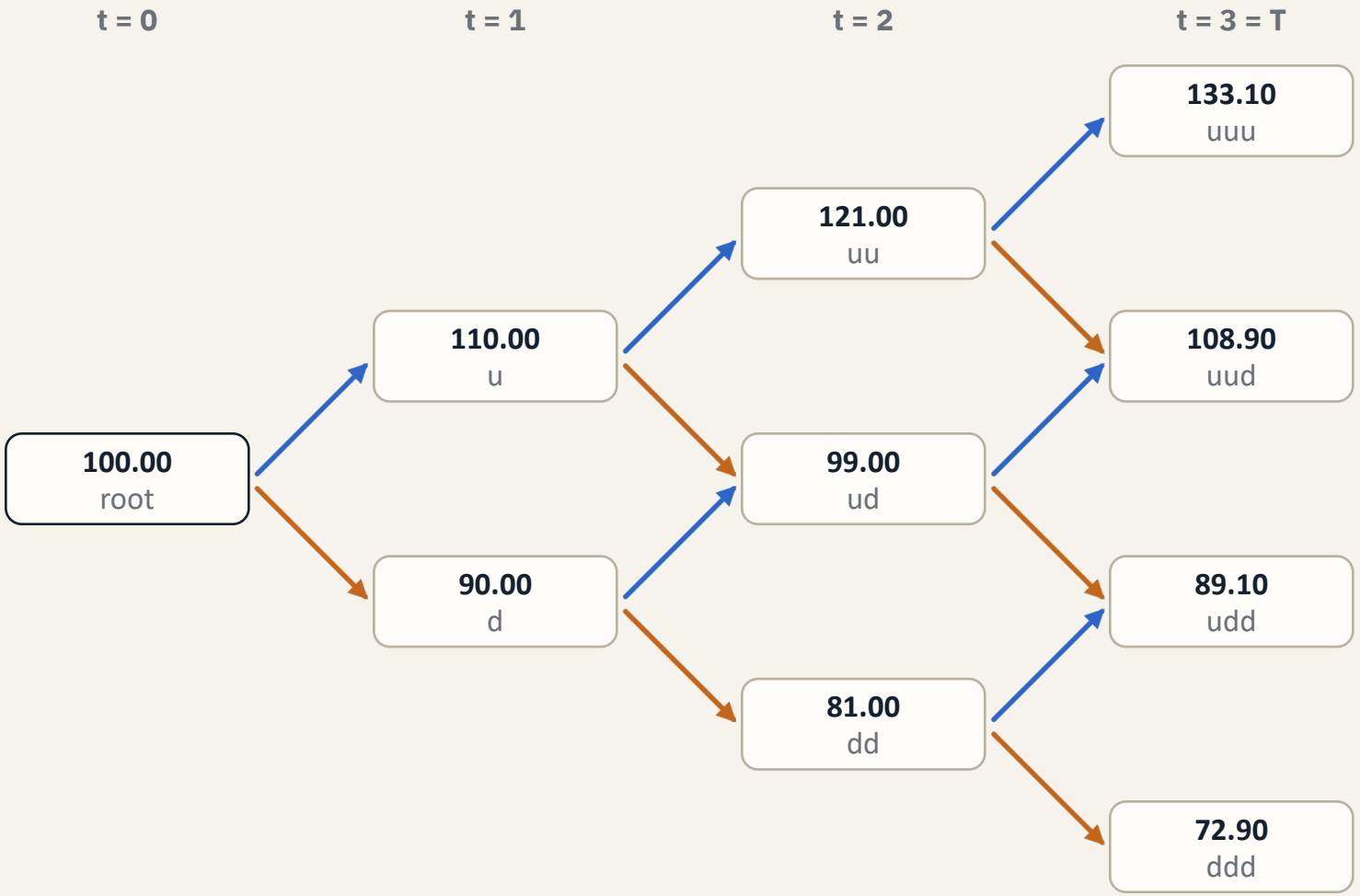


# Forward pass ①: propagate prices

Start at the root and push prices to the right. Each node feeds two children:

$S\uparrow = S \cdot 1.10$
$S\downarrow = S \cdot 0.90$

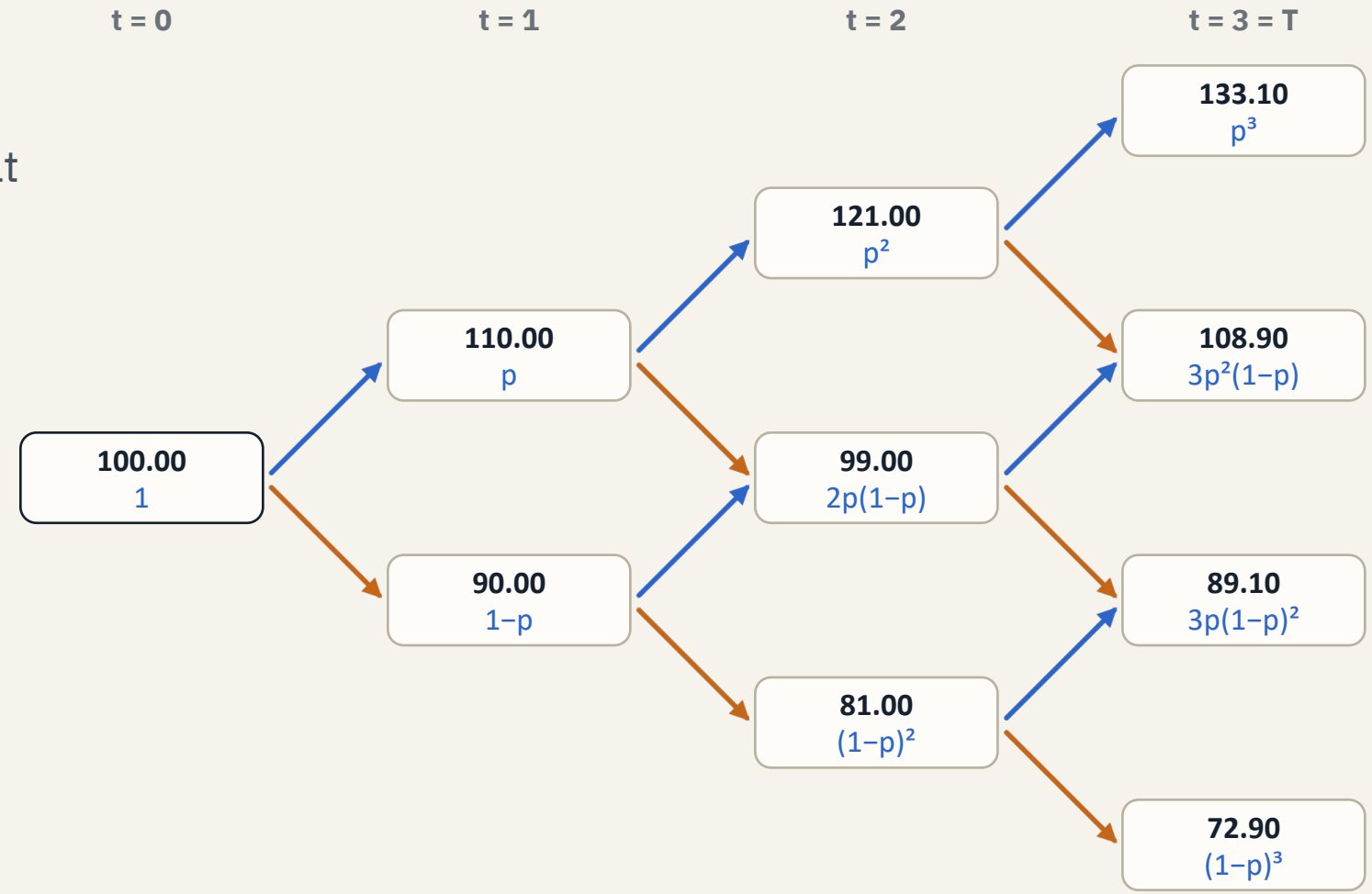
Purely mechanical — no probabilities yet. The leaves span every possible outcome: **72.90 … 133.10**.



# Forward pass ②: propagate probabilities

Probability mass flows right: a node sends a share p up and 1 – p down; merging paths add up. The result is the **binomial distribution** — Pascal's triangle × $p^j(1-p)^{n-j}$.

| S(T) | p = .7 | q = .6 |
|---|---|---|
| 133.10 | 34.3% | 21.6% |
| 108.90 | 44.1% | 43.2% |
| 89.10 | 18.9% | 28.8% |
| 72.90 | 2.7% | 6.4% |

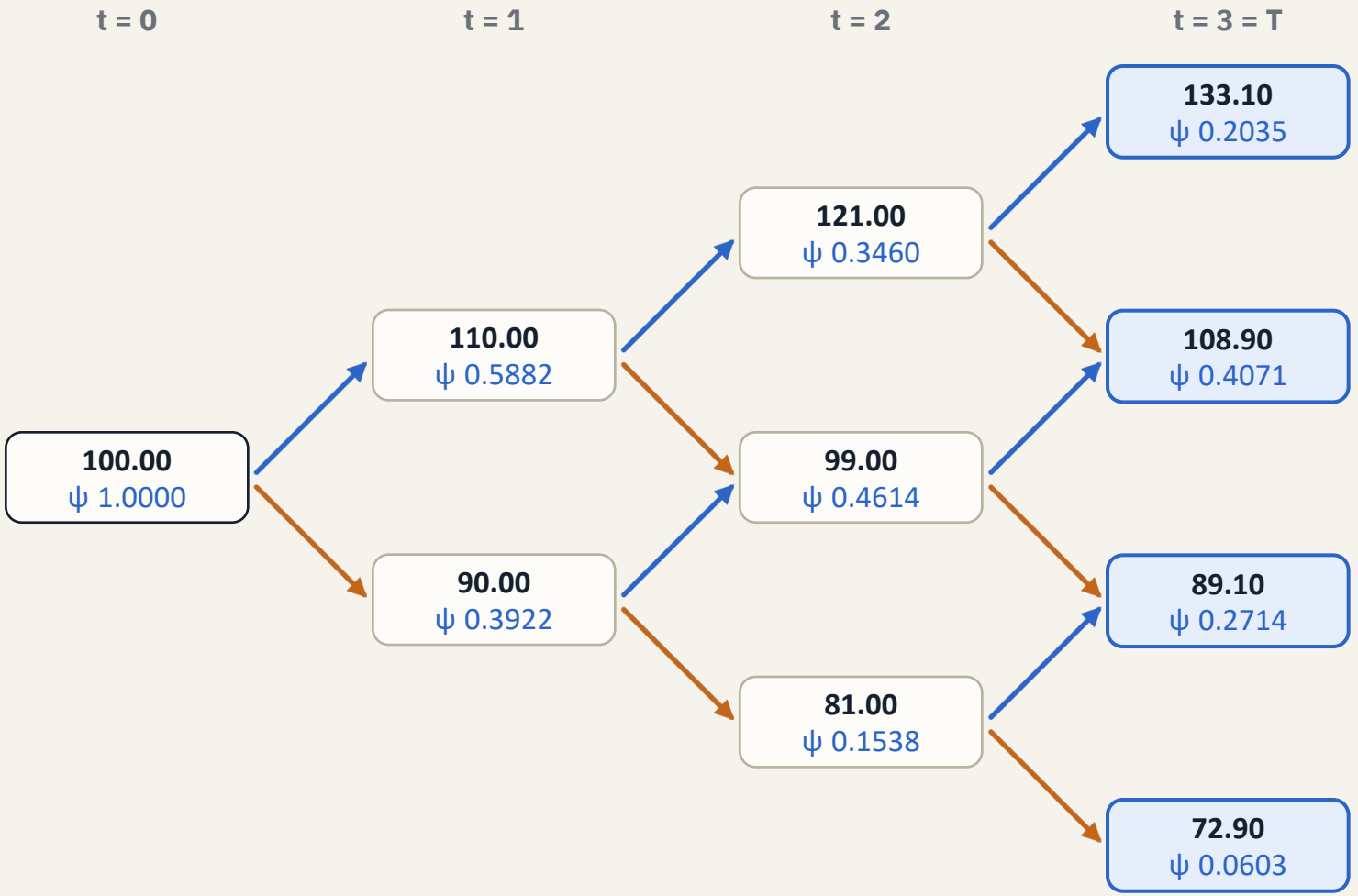


# Forward pass ③: propagate state prices

**ψ(node)** = what you would pay today for $1 delivered only if the stock lands on that node (Arrow-Debreu price).

$$\psi(\text{child}) = \Sigma\, \psi(\text{parent}) \cdot w / R$$
$w = q$ (up) or $1 - q$ (down)

Same flow as the probabilities, but with q and a discount per step. The leaves sum to **$1/R^3 = 0.9423$**: a sure $1 at T.

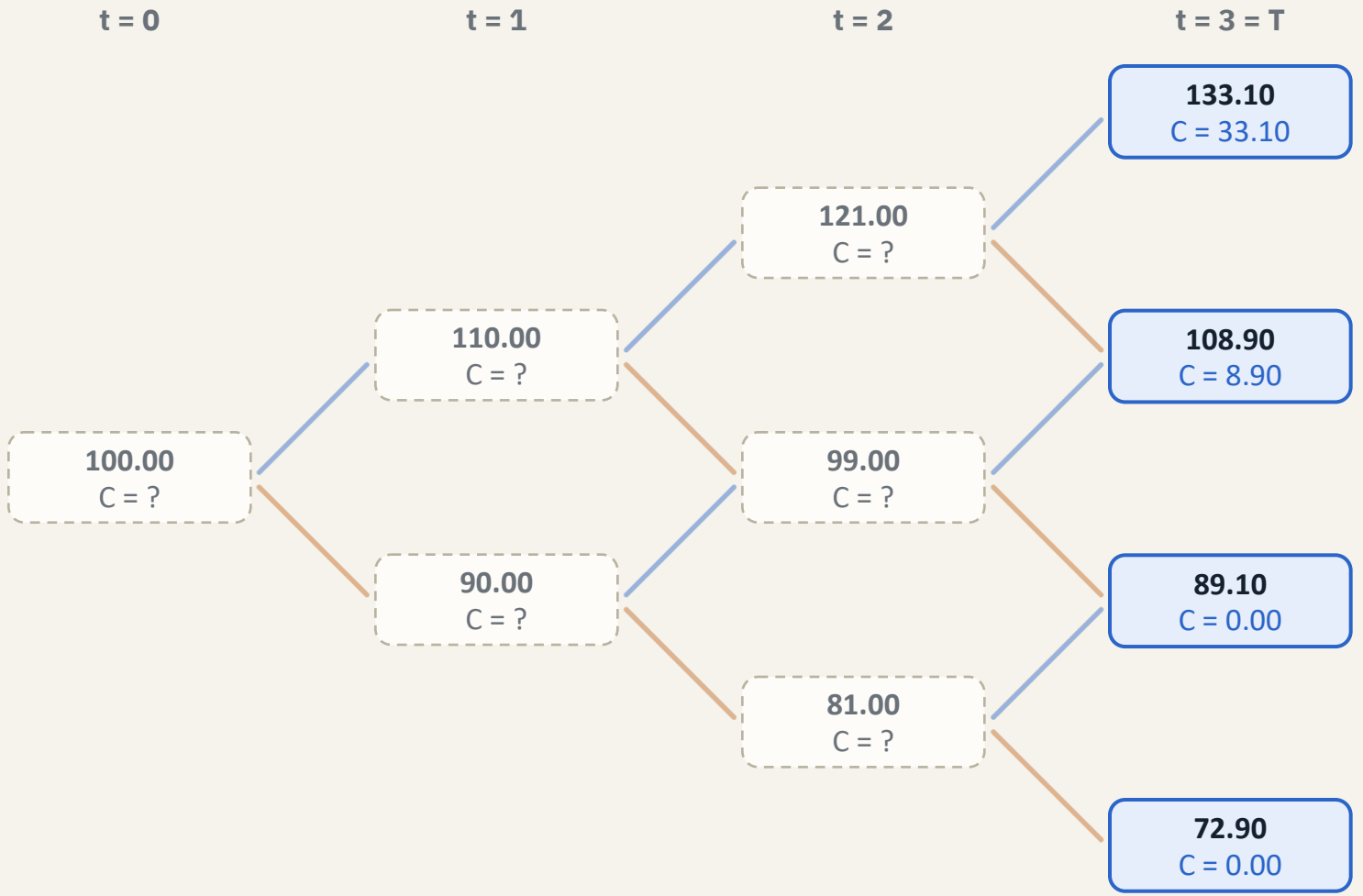


# At expiry, the value is simply the payoff

On the leaves there is no uncertainty left. For a call with K = 100:

$$C(T) = \max(S - 100, 0)$$

Every earlier node is still a question mark. To fill them in we turn around and walk **backward in time**.



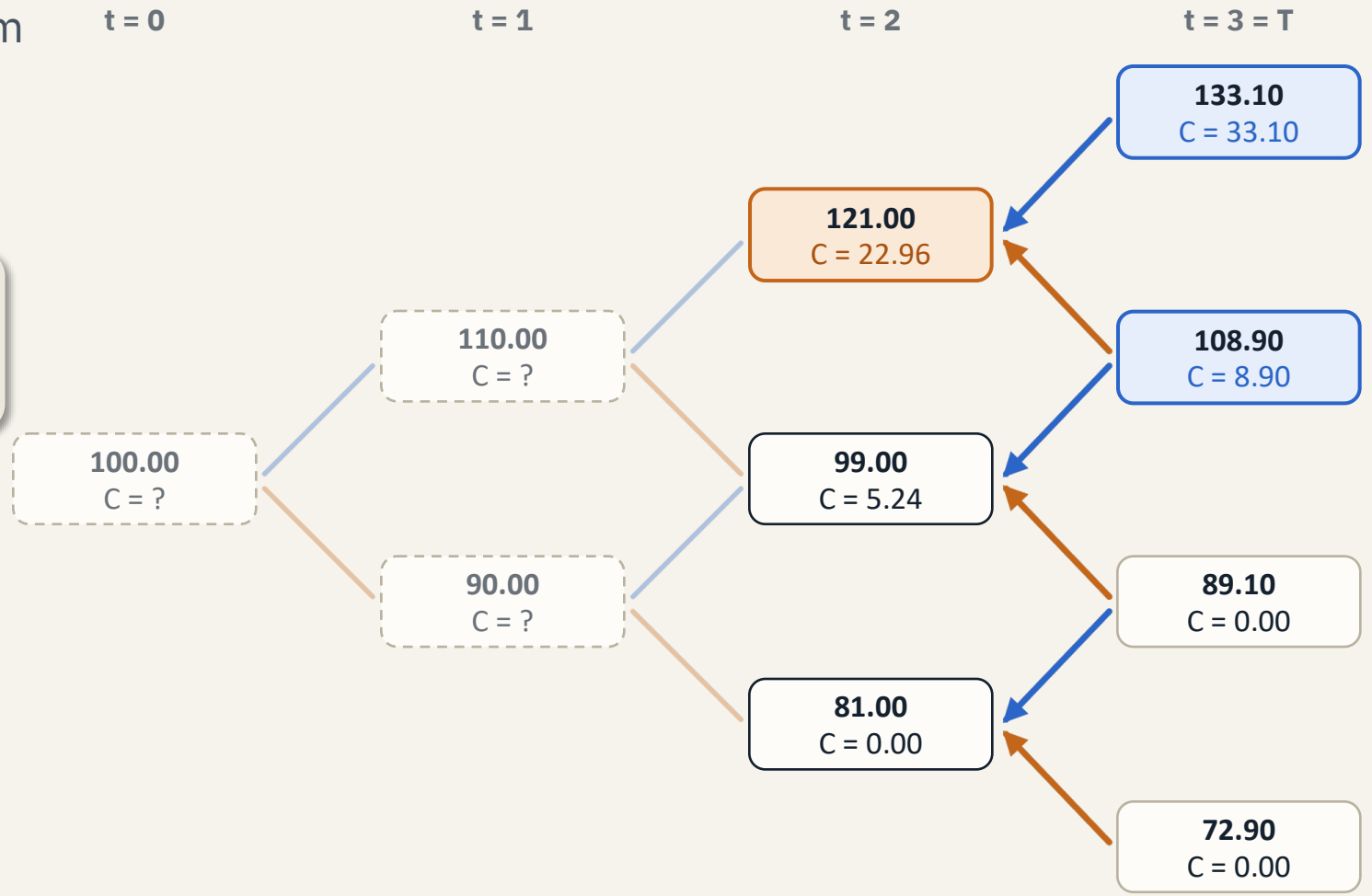
# Backward pass: one step at a time

Every node is a one-step problem we already solved. Apply the same rule everywhere:

$$V = [q \cdot V\uparrow + (1-q) \cdot V\downarrow] / R$$

Top node at t = 2:

(0.6·33.10 + 0.4·8.90) / 1.02 = **22.96**

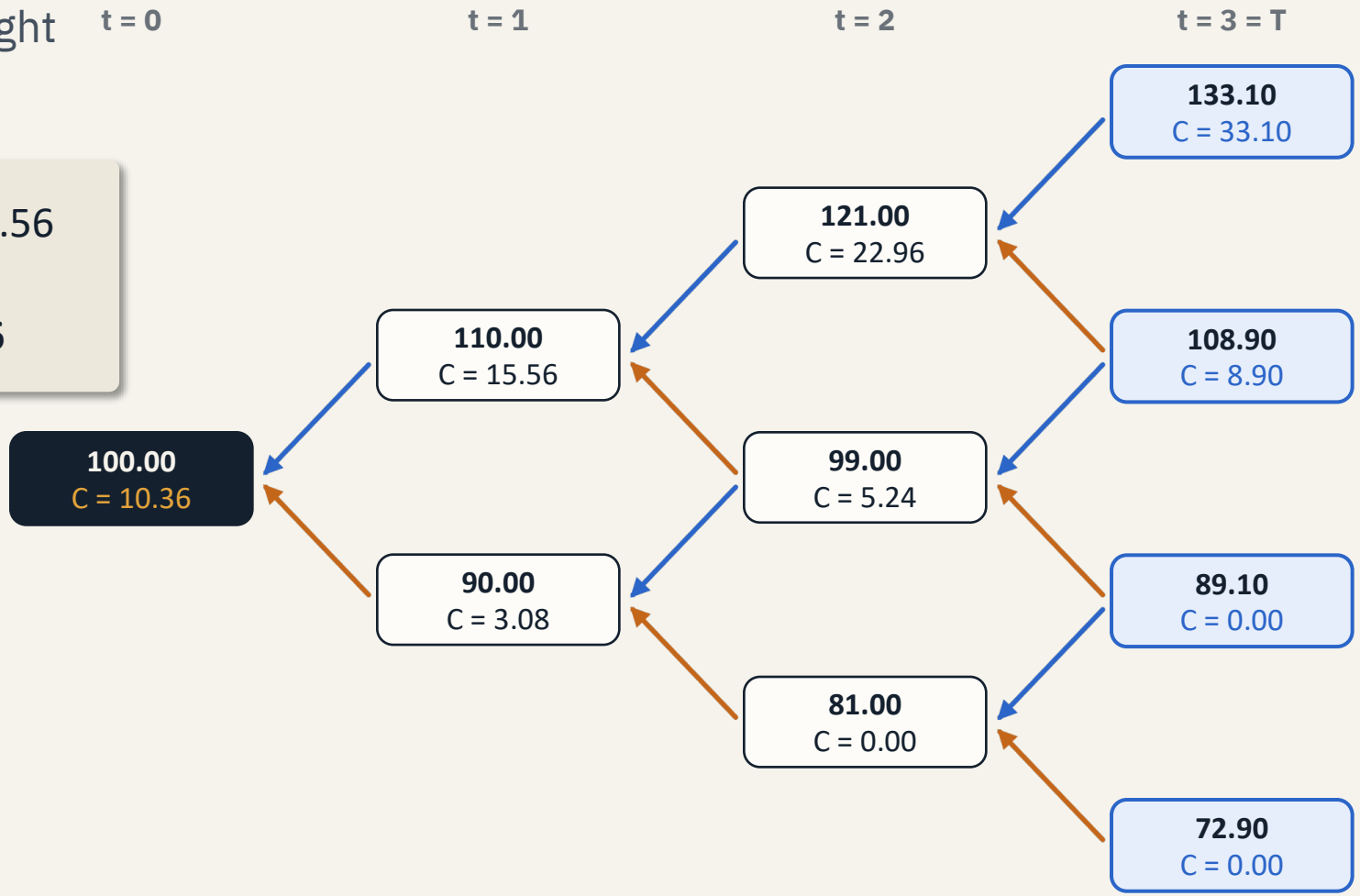


# Repeat to the root: the option is worth 10.36

Same rule, column by column, right to left:

t=1 ↑ (.6·22.96+.4·5.24)/1.02 = 15.56
t=1 ↓ (.6·5.24+.4·0)/1.02 = 3.08
t=0 (.6·15.56+.4·3.08)/1.02 = **10.36**

Ten nodes, nine multiply-adds. Real-world p was never used.



# Two routes through the lattice, one price

## Forward → state prices · payoff

| Leaf S | payoff | ψ | ψ·payoff |
|---|---|---|---|
| 133.10 | 33.10 | 0.2035 | 6.737 |
| 108.90 | 8.90 | 0.4071 | 3.623 |
| 89.10 | 0.00 | 0.2714 | 0.000 |
| 72.90 | 0.00 | 0.0603 | 0.000 |
| Σ | | | 10.360 |

## Backward ← induction

| Step | Values, top to bottom |
|---|---|
| t = 3 | 33.10 · 8.90 · 0 · 0 |
| t = 2 | 22.96 · 5.24 · 0 |
| t = 1 | 15.56 · 3.08 |
| t = 0 | $C_0$ = 10.360 |

$$C_0 = R^{-n} \cdot \Sigma_j \, (n \text{ choose } j) \cdot q^j (1-q)^{n-j} \cdot \text{payoff}(S_0 \cdot u^j \cdot d^{n-j})$$

Forward sums ψ against payoffs; backward nests the same sum. Same numbers, opposite order of evaluation.

# The whole method fits on one screen

```
def price(S0, K, r, sigma, T, n, call=True, american=False):
    dt = T / n
    u = exp(sigma * sqrt(dt)); d = 1 / u
    R = exp(r * dt); q = (R - d) / (u - d)
    # forward: prices at the leaves
    S = [S0 * u**j * d**(n - j) for j in range(n + 1)]
    V = [max(s - K, 0) if call else max(K - s, 0) for s in S]
    # backward: roll values to the root
    for i in range(n - 1, -1, -1):
        for j in range(i + 1):
            V[j] = (q * V[j + 1] + (1 - q) * V[j]) / R
            if american:
                s = S0 * u**j * d**(i - j)
                V[j] = max(V[j], (s - K) if call else (K - s))
    return V[0]
```

**O(n²) time**
n(n+1)/2 node updates

**O(n) memory**
One array, overwritten in place

**n = 1000 →**
call 10.4486
BS 10.4506

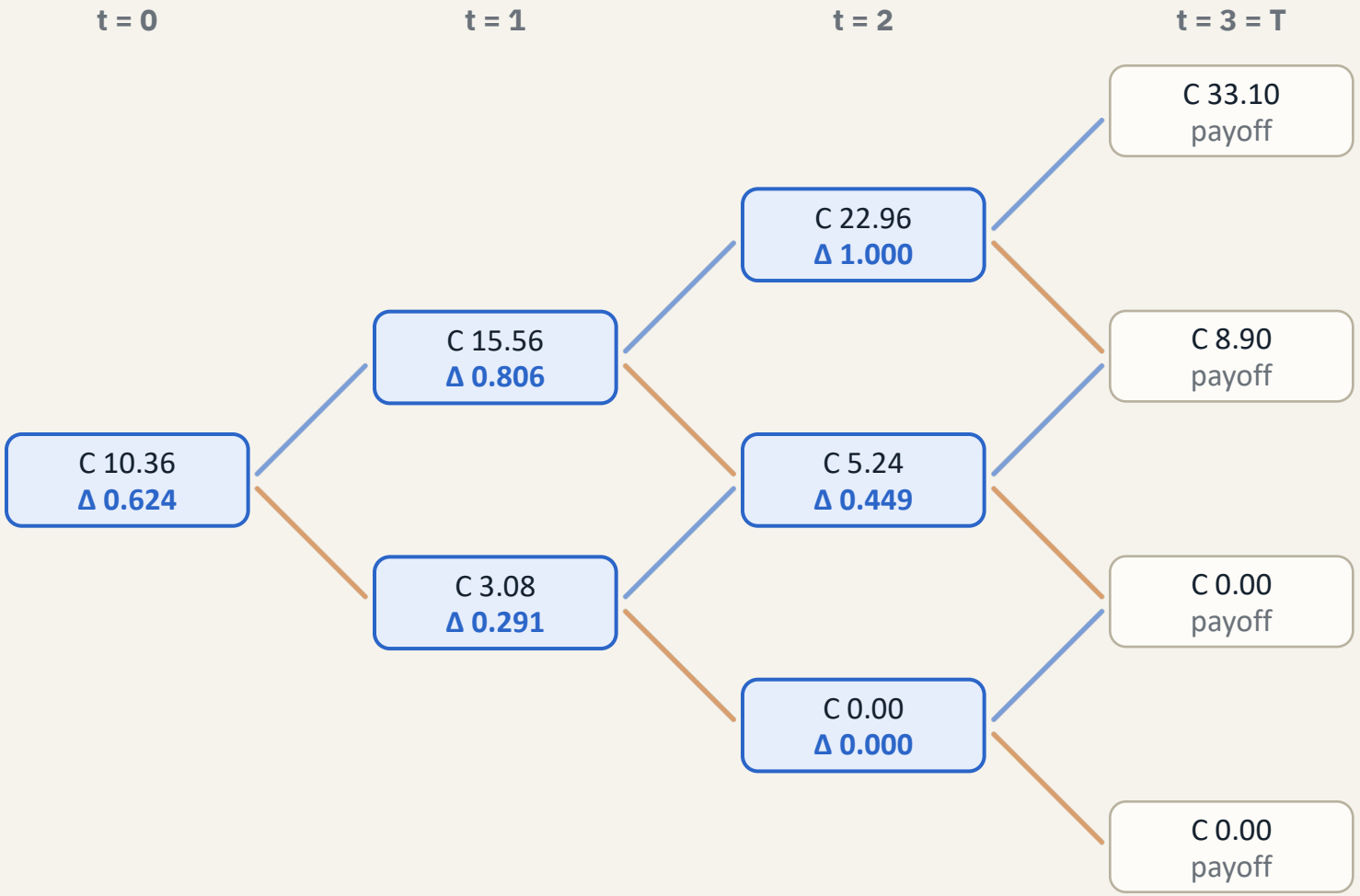
# Every node carries its own hedge

At each node, hold the one-step clone:

$$\Delta = (V\uparrow - V\downarrow)/(S\uparrow - S\downarrow)$$
$$B = V - \Delta \cdot S$$

Today: 0.624 shares (62.40) and borrow 52.04 → 10.36. Rebalance after each move; the portfolio always equals the node value, with **no cash added or removed**.

Δ → 1 deep in the money, Δ → 0 far out of it.



# The Greeks: a risk dashboard for one option

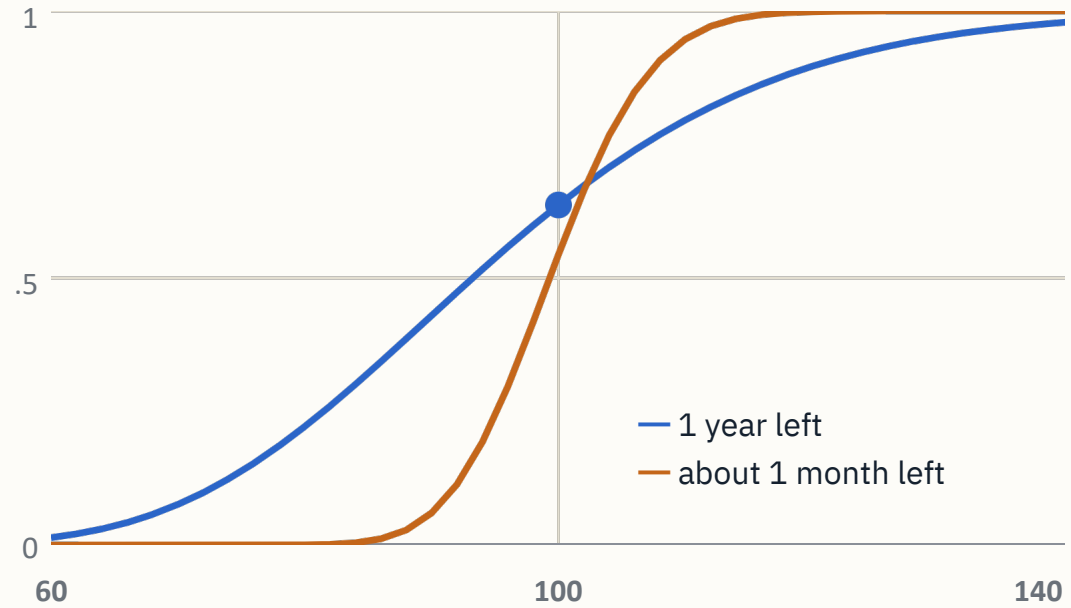
A price is one number. A trader also needs to know **how that number will move**. Each Greek = the change in V when one input moves, with all others frozen.

| Greek | Sensitivity to | The question it answers | Our call* |
|---|---|---|---|
| **Δ** Delta | stock price S | Stock +1 → option +how much? | +0.64 |
| **Γ** Gamma | delta, as S moves | How fast does my hedge go stale? | +0.019 |
| **Θ** Theta | time passing | What does one more day cost me? | −0.018/day |
| **ν** Vega | volatility σ | Market gets jumpier by 1 point → ? | +0.38 |
| **ρ** Rho | interest rate r | Rates up by 1 point → ? | +0.53 |

</> Developer view: the Greeks are the **gradient** of the pricing function (plus one second derivative) — what autodiff computes in machine learning.

* European call, S = K = 100, r = 5%, σ = 20%, T = 1 year; value 10.45

# Delta: speed, hedge ratio, and a hint of probability




**1 · Speed**

Δ = 0.64: stock +1.00 → call +0.64. Like a car's speed: how fast V moves with S.

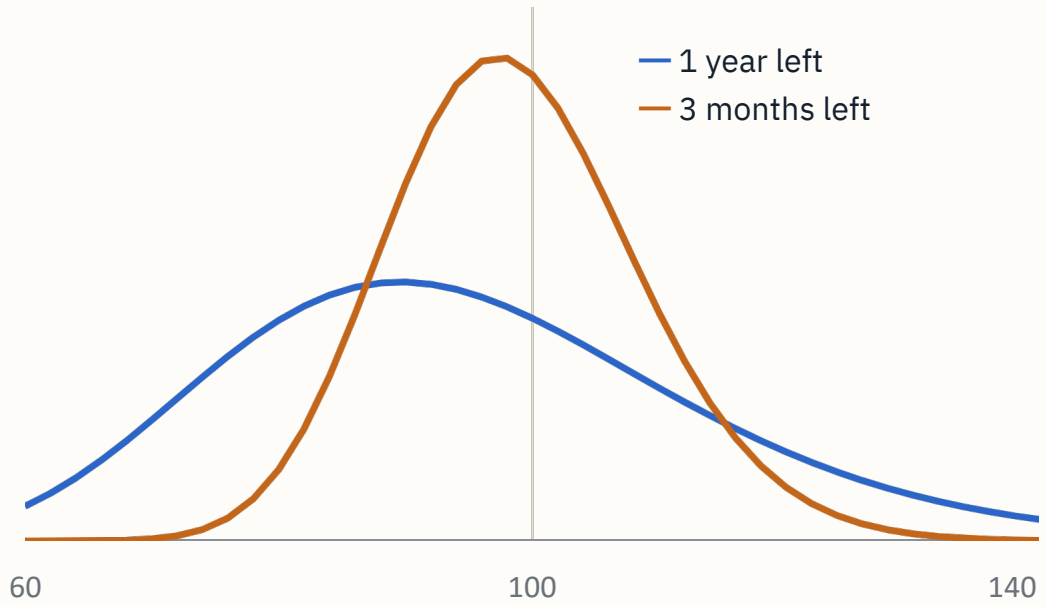
**2 · Hedge ratio**

Sold 100 calls? Buy 64 shares and small moves cancel. The tree's root said 0.624.

**3 · "How in the money"**

Δ → 1: behaves like the stock. Δ → 0: a lottery ticket. Near expiry it becomes a step: all or nothing.

# Gamma: acceleration — and why holders love big moves




stock price S · call gamma (peaks at the strike K = 100)

### 1 · Acceleration

Γ = 0.019: stock +1 → delta +0.019. In the tree: Δ 0.29 at S = 90, 0.81 at S = 110.

### 2 · Hedging workload

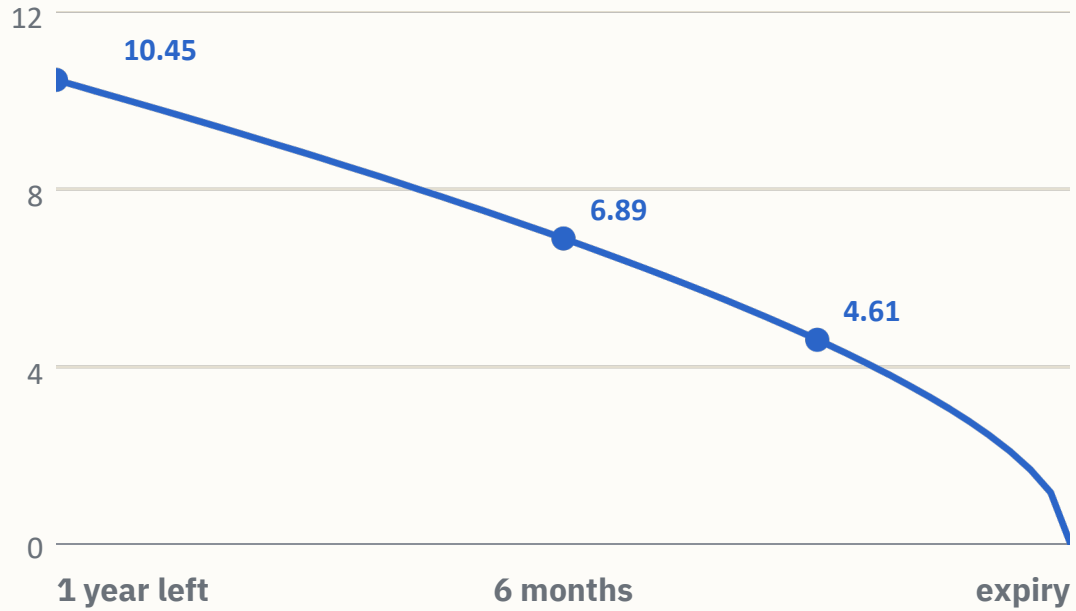
High gamma = the hedge goes stale fast = rebalance often. Worst at the strike, near expiry.

### 3 · Convexity pays

gain ≈ ½ · Γ · (move)²

Squared: up or down, a hedged holder gains from any move.

# Theta: the melting ice cube — rent paid for gamma




ATM call value, stock held at 100: 10.45 → 6.89 → 4.61 → 0

### 1 · Time decay

Θ = −6.41 per year ≈ **−0.018 per day**: the cost of waiting, all else equal.

### 2 · Accelerates at the end

Less time = less chance to move. The last weeks melt fastest (1.18 with a week left).

### 3 · No free lunch

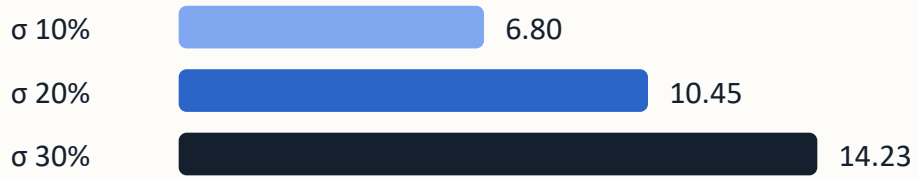
$$\Theta + \tfrac{1}{2}\sigma^2 S^2 \Gamma + rS\Delta = rV$$

The PDE again: gamma gains are paid for by theta, day by day.

# Vega and rho: sensitivity to the market’s mood and to rates

## Vega · +0.38 per vol point

More uncertainty = more valuable. Upside is kept, downside is capped, so a wider spread only helps the holder.




Run it backwards: from a market price, solve for σ = **implied volatility**, the market’s “fear gauge” (VIX).

## Rho · +0.53 per rate point

A call lets you pay K later instead of buying the stock now; higher rates make paying later worth more. A put’s rho is negative.

Usually the smallest Greek for short-dated equity options, but large for long-dated and rate options.

### ON THE LATTICE: BUMP AND REVALUE

```
vega = (price(σ=.21) − price(σ=.19))/2 # 0.375
rho  = (price(r=.06) − price(r=.04))/2 # 0.532
```

# On the lattice: Δ, Γ, Θ for free from the nodes

## DELTA · ∂V/∂S

**0.624**

$(C\uparrow - C\downarrow)/(S\uparrow - S\downarrow)$
$= (15.56 - 3.08)/20$

Slope across the t = 1 nodes

## GAMMA · ∂²V/∂S²

**0.026**

$(\Delta\uparrow - \Delta\downarrow)/((S\uparrow\uparrow - S\downarrow\downarrow)/2)$
$= (0.806 - 0.291)/20$

Change in slope across t = 2

## THETA · ∂V/∂T

**−2.56**

$(C(ud) - C_0)/(2\ \text{steps})$
$= (5.24 - 10.36)/2$

Per step; exact when d = 1/u

**Vega and rho** aren't on the tree: bump σ or r, rebuild, and difference the two prices.

# Putting it together: the Greeks explain the P&L

$$\text{change in } V \approx \Delta \cdot dS + \tfrac{1}{2}\Gamma \cdot dS^2 + \Theta \cdot dt + \nu \cdot d\sigma + \rho \cdot dr$$

A Taylor expansion. Example: one day passes and the stock moves 100 → 102.

| Term | Calculation | Contribution |
|---|---|---|
| Delta | 0.637 × 2 | +1.274 |
| Gamma | ½ × 0.0188 × 2² | +0.038 |
| Theta | −6.41 × 1/365 | −0.018 |
| Greeks' estimate | | +1.294 |
| Full repricing | | +1.293 |

Bank risk reports are this table, summed over thousands of positions: **a linearised model of the portfolio**, refreshed many times a day.

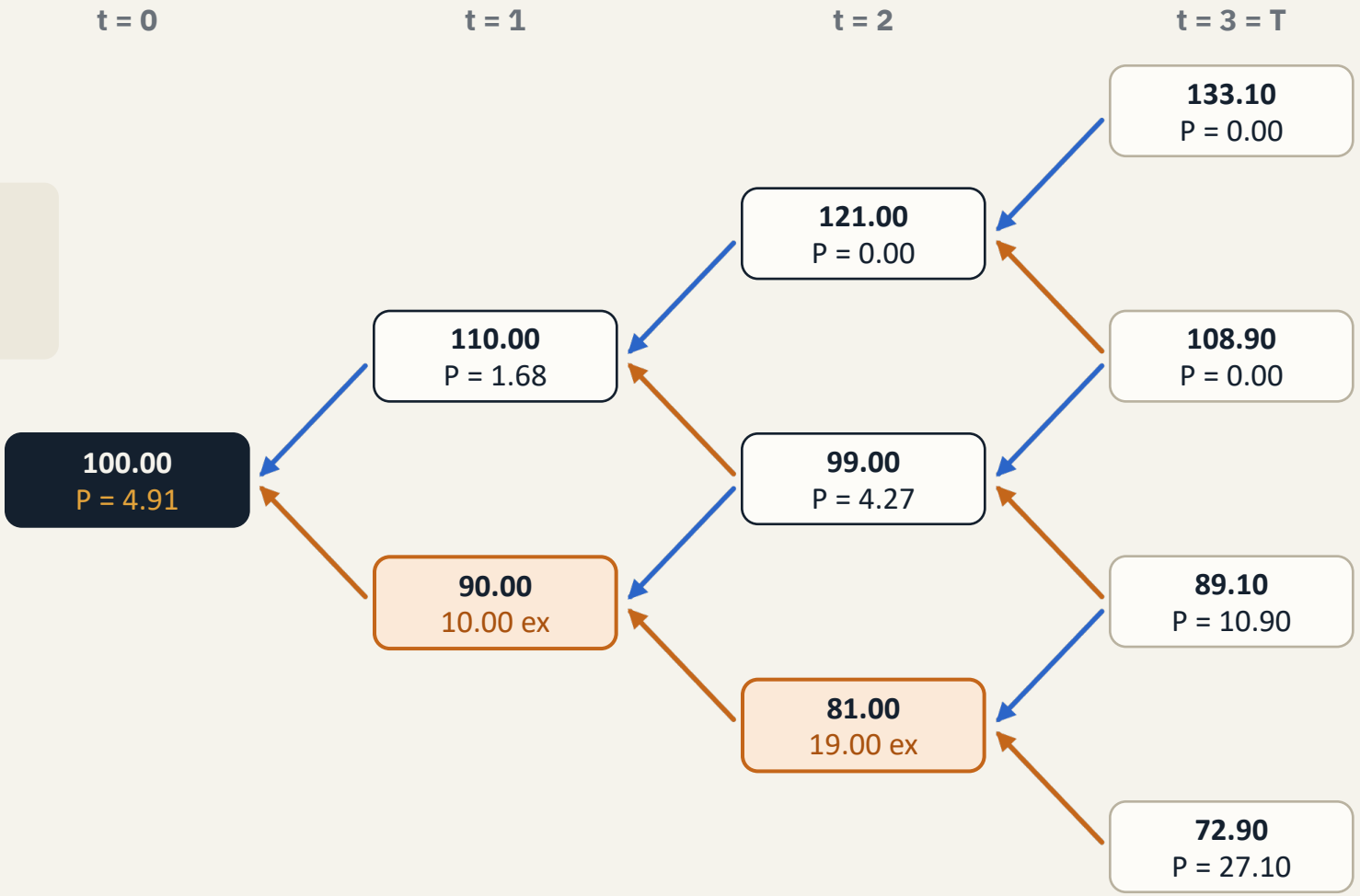
# American put: exercise or wait, at every node

The holder may exercise any time. The backward step gains one **max**:

$$V = \max( K - S, [q \cdot V\uparrow + (1-q) \cdot V\downarrow]/R )$$

exercise vs continue

Put, K = 100. At S = 81: exercise **19.00** beats continue 17.04. At S = 90: **10.00** beats 9.97.



# The right to choose has a price

EUROPEAN PUT

**4.59**

Same tree, no max

AMERICAN PUT

**4.91**

Early-exercise premium +0.32

AMERICAN CALL

**10.36**

= European: never exercise early without dividends

## Sanity check: put-call parity

$C - P = S_0 - K/R^3$

$10.36 - 4.59 = 100 - 94.23 = 5.77$ ✓

## Realistic tree, n = 1000

S=K=100, r=5%, σ=20%, T=1

European 5.57 · American 6.09

# Calibrate the lattice to σ and r (Cox-Ross-Rubinstein)

TIME STEP

$\Delta t = T / n$

More steps, finer grid in time.

MOVE SIZES

$u = e^{(\sigma\sqrt{\Delta t})}$

$d = 1/u$

Log-return variance per step = $\sigma^2\Delta t$. The tree recombines around $S_0$.

RISK-NEUTRAL WEIGHT

$q = (e^{(r\Delta t)} - d) / (u - d)$

Expected growth per step is exactly $e^{(r\Delta t)}$.

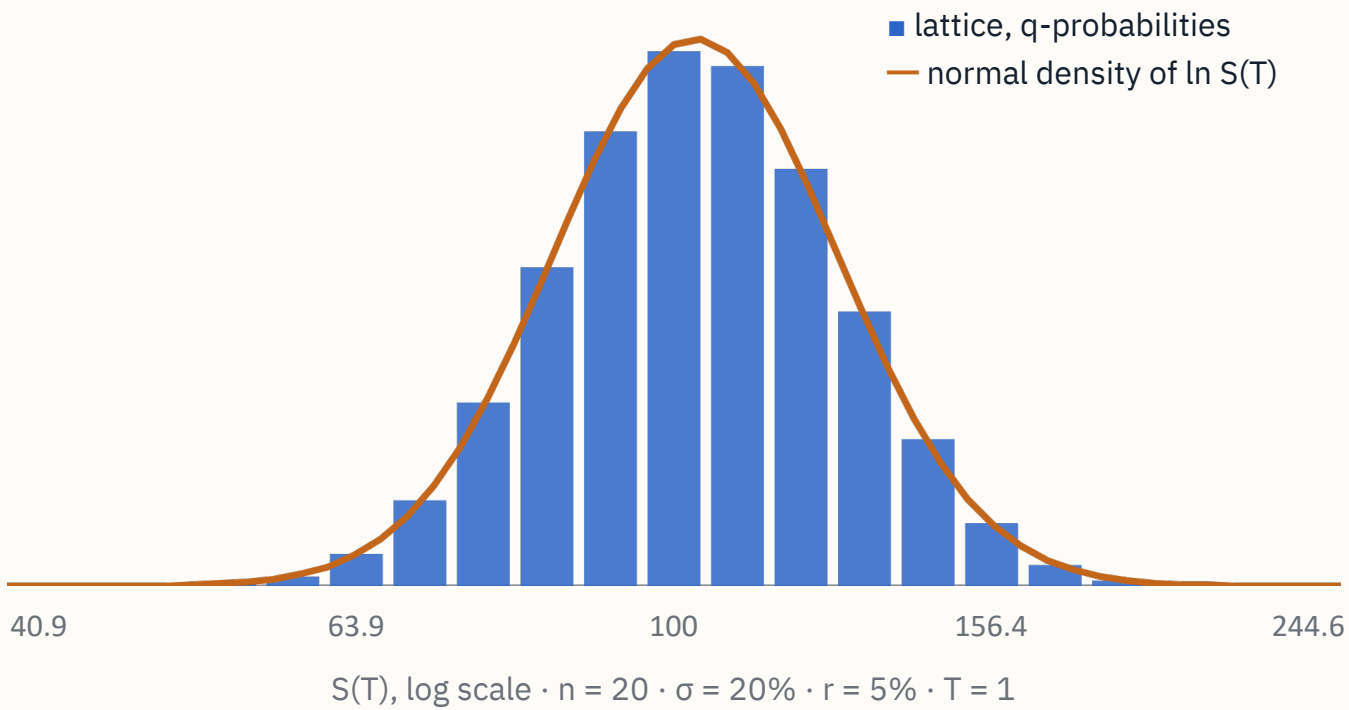
**$\sigma = 20\%, r = 5\%, T = 1, n = 4$ : $\Delta t = 0.25 \cdot u = 1.1052 \cdot d = 0.9048 \cdot q = 0.5378$**

# The backward step *is* the Black-Scholes PDE, discretised

| | | |
|---|---|---|
| 1 | $R \cdot V(t, S) = q \cdot V(t+\Delta t, uS) + (1-q) \cdot V(t+\Delta t, dS)$ | the lattice rule |
| 2 | Taylor-expand, using $E_q[\Delta S] \approx rS\Delta t$ and $E_q[\Delta S^2] \approx \sigma^2 S^2 \Delta t$:<br>$RHS \approx V + (\partial V/\partial t + rS \cdot \partial V/\partial S + \frac{1}{2}\sigma^2 S^2 \cdot \partial^2 V/\partial S^2) \cdot \Delta t$ | first two moments of one step under q |
| 3 | LHS ≈ V + rV·Δt. Cancel V, divide by Δt, let Δt → 0: | R = e^(rΔt) ≈ 1 + rΔt |

$$\partial V/\partial t + r \cdot S \cdot \partial V/\partial S + \frac{1}{2} \cdot \sigma^2 \cdot S^2 \cdot \partial^2 V/\partial S^2 - r \cdot V = 0$$

# The forward pass converges to the lognormal




Forward-propagating q gives a **binomial** count of up-moves at T.

By the central limit theorem, ln S(T) becomes **normal**: mean $\ln S_0 + (r - \sigma^2/2)T$, variance $\sigma^2 T$.

That is geometric Brownian motion under Q — Black-Scholes' assumption.

# The lattice sum splits into the two Black-Scholes terms

**BINOMIAL CALL PRICE · A = FEWEST UP-MOVES THAT FINISH ABOVE K**

$$C_0 = S_0 \cdot P[\mathrm{Bin}(n, q') \geq a] - K \cdot R^{-n} \cdot P[\mathrm{Bin}(n, q) \geq a],\ q' = q \cdot u / R$$



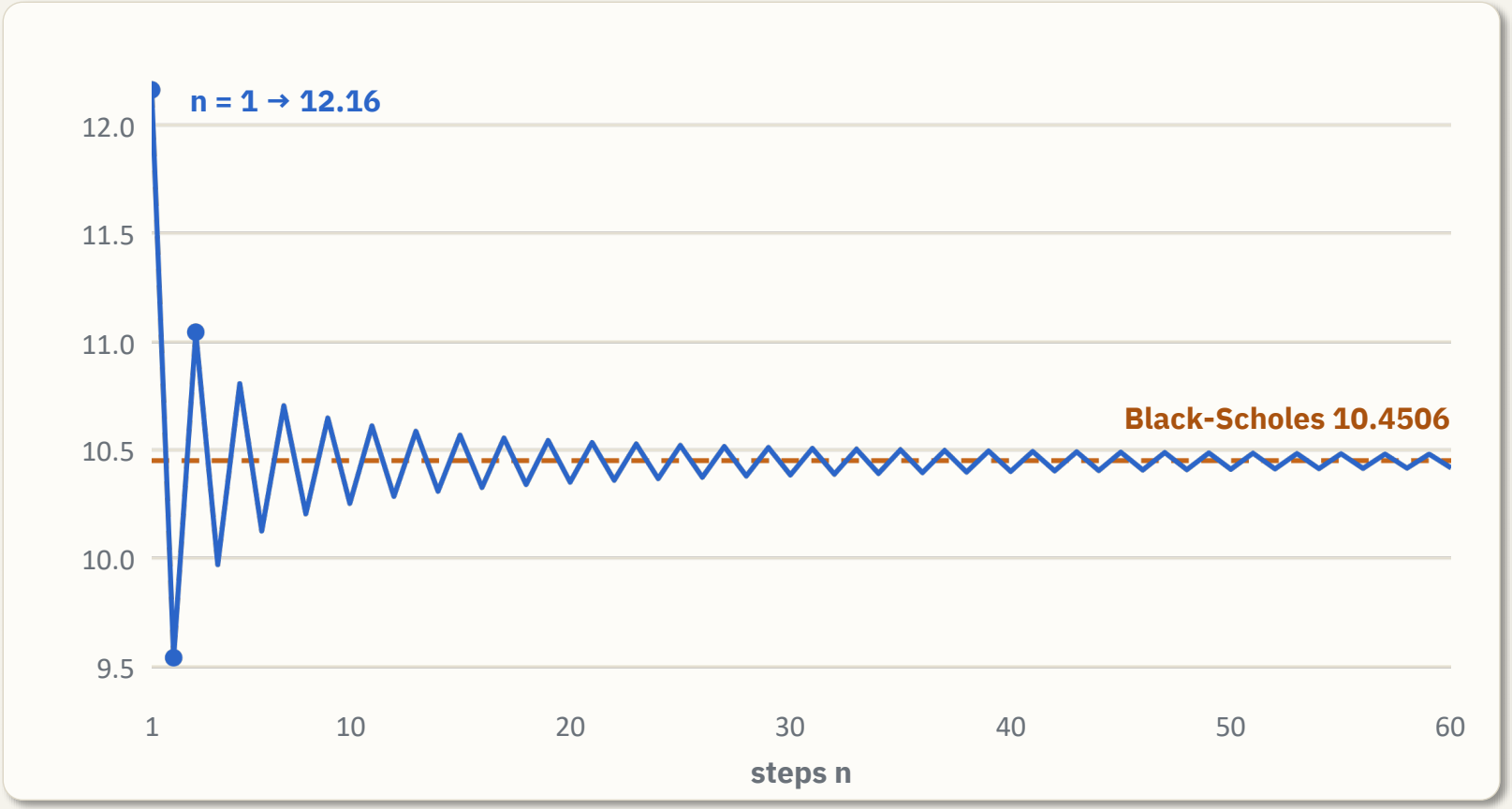
$n \to \infty$ (CLT on each binomial)

$$C = S \cdot N(d_1) - K \cdot e^{(-rT)} \cdot N(d_2)$$

$N(d_2)$: risk-neutral chance of finishing in the money. $N(d_1)$: the same with the stock as unit of account — and the hedge Δ.

Toy tree check (n = 3, a = 2): 100·0.7142 − 94.23·0.6480 = 71.42 − 61.06 = **10.36** ✓

# More steps, closer to 10.4506




| n | price |
|---|---|
| 1 | 12.1623 |
| 10 | 10.2534 |
| 100 | 10.4306 |
| 1000 | 10.4486 |
| BS | 10.4506 |

Error shrinks roughly like 1/n. The odd/even zig-zag comes from where K falls between leaves.

# A dictionary: lattice ↔ Black-Scholes

| Concept | Lattice (discrete) | Black-Scholes (continuous) |
|---|---|---|
| Time | n steps of $\Delta t = T/n$ | continuous t, $dt \to 0$ |
| Randomness | $\times u$ or $\times d$, $u = e^{\sigma\sqrt{\Delta t}}$ | $dS = rS\cdot dt + \sigma S\cdot dW$ |
| Pricing weights | $q = (R - d)/(u - d)$ | risk-neutral measure Q, drift r |
| Forward pass | binomial probabilities, state prices $\psi$ | lognormal density, Fokker-Planck |
| Backward pass | $V = [q\cdot V\uparrow + (1-q)\cdot V\downarrow]/R$ | BS PDE, solved from payoff at T |
| Hedge | $\Delta = (V\uparrow - V\downarrow)/(S\uparrow - S\downarrow)$ | $\Delta = \partial V/\partial S = N(d_1)$ for a call |
| Price | binomial sum over the leaves | $S\cdot N(d_1) - K\cdot e^{-rT}\cdot N(d_2)$ |
| Early exercise | max(exercise, continue) per node | free-boundary problem, no closed form |

TAKEAWAYS

# Five things to remember

1 **Price by replication, not prediction.** The real-world p never enters.

2 **q = (R − d)/(u − d) is a pricing device.** A world where everything grows at r.

3 **Forward builds the stage, backward computes value.** Prices, probabilities, state prices →; values ←.

4 **Nodes give hedges and Greeks; one max gives American options.**

5 **Shrink Δt and the lattice is Black-Scholes.** Backward step → PDE; binomial → lognormal; sum → $N(d_1)$, $N(d_2)$.

A FINAL WORD

# Options changed how the world manages risk

Options trading gave businesses and investors more ways to protect against adverse price moves, supporting economic activity under uncertainty.

Leveraged positions can also magnify losses and transmit stress through financial markets.

**— The End —**

